

Chapter 4 Exercises: Section 4.1

SOLUTIONS TO RICK DURRETT'S
Probability: Theory and Examples

June 12, 2026

These solutions use the local Chapter 4 LLM/Viki knowledge set in `Probability/LLM_Viki_Dataset/Chapter4_Knowledge`. The cited knowledge IDs refer to entries in `chapter4_knowledge_pieces.jsonl`.

Exercise 4.1.1

Question. Let A be an event with $\mathbb{P}(A) > 0$ and let $G \in \mathcal{G}$. Prove the conditional Bayes formula

$$\mathbb{P}(G \mid A) = \frac{\int_G \mathbb{P}(A \mid \mathcal{G}) d\mathbb{P}}{\int_{\Omega} \mathbb{P}(A \mid \mathcal{G}) d\mathbb{P}}.$$

When $\mathcal{G}$ is generated by a countable partition, identify the usual Bayes formula as a special case.

Knowledge used. `durrett_ch4_conditional_expectation_definition`;
`durrett_ch4_partition_conditioning_formula`;
`durrett_ch4_conditional_expectation_properties`.

Solution. By definition, $\mathbb{P}(A \mid \mathcal{G}) = \mathbb{E}(\mathbf{1}_A \mid \mathcal{G})$. Since $G \in \mathcal{G}$, the defining property of conditional expectation gives

$$\int_G \mathbb{P}(A \mid \mathcal{G}) d\mathbb{P} = \int_G \mathbb{E}(\mathbf{1}_A \mid \mathcal{G}) d\mathbb{P} = \int_G \mathbf{1}_A d\mathbb{P} = \mathbb{P}(A \cap G).$$

Taking $G = \Omega$ gives

$$\int_{\Omega} \mathbb{P}(A \mid \mathcal{G}) d\mathbb{P} = \mathbb{P}(A).$$

Therefore

$$\frac{\int_G \mathbb{P}(A \mid \mathcal{G}) d\mathbb{P}}{\int_{\Omega} \mathbb{P}(A \mid \mathcal{G}) d\mathbb{P}} = \frac{\mathbb{P}(A \cap G)}{\mathbb{P}(A)} = \mathbb{P}(G \mid A).$$

If $\mathcal{G} = \sigma(G_1, G_2, \dots)$ is generated by a countable partition with $\mathbb{P}(G_i) > 0$, then $\mathbb{P}(A \mid \mathcal{G})$ is constant on each G_i and equals $\mathbb{P}(A \mid G_i)$. Hence, for the atom G_i ,

$$\mathbb{P}(G_i \mid A) = \frac{\mathbb{P}(A \mid G_i)\mathbb{P}(G_i)}{\sum_j \mathbb{P}(A \mid G_j)\mathbb{P}(G_j)}.$$

Exercise 4.1.2

Question. Prove the conditional Chebyshev inequality: for $a > 0$,

$$\mathbb{P}(|X| \geq a \mid \mathcal{F}) \leq a^{-2} \mathbb{E}(X^2 \mid \mathcal{F}).$$

Knowledge used. `durrett_ch4_conditional_expectation_definition`;
`durrett_ch4_conditional_expectation_properties`.

Solution. Pointwise,

$$\mathbf{1}_{\{|X| \geq a\}} \leq \frac{X^2}{a^2}.$$

Taking conditional expectations with respect to $\mathcal{F}$ and using monotonicity gives

$$\mathbb{P}(|X| \geq a \mid \mathcal{F}) = \mathbb{E}(\mathbf{1}_{\{|X| \geq a\}} \mid \mathcal{F}) \leq \mathbb{E}\left(\frac{X^2}{a^2} \mid \mathcal{F}\right) = a^{-2} \mathbb{E}(X^2 \mid \mathcal{F}).$$

Exercise 4.1.3

Question. Prove the conditional Cauchy-Schwarz inequality

$$\mathbb{E}(XY \mid \mathcal{G})^2 \leq \mathbb{E}(X^2 \mid \mathcal{G}) \mathbb{E}(Y^2 \mid \mathcal{G}).$$

Knowledge used. `durrett_ch4_conditional_expectation_properties`;
`durrett_ch4_taking_out_what_is_known`; `durrett_ch4_conditional_expectation_l2_projection`.

Solution. Assume $X, Y \in L^2$. Put

$$A = \mathbb{E}(XY \mid \mathcal{G}), \quad B = \mathbb{E}(Y^2 \mid \mathcal{G}), \quad C = \mathbb{E}(X^2 \mid \mathcal{G}).$$

On the event $\{B = 0\} \in \mathcal{G}$, we have

$$\int_{\{B=0\}} Y^2 d\mathbb{P} = \int_{\{B=0\}} B d\mathbb{P} = 0,$$

so $Y = 0$ a.s. on $\{B = 0\}$. Thus $A = 0$ a.s. on $\{B = 0\}$.

On $\{B > 0\}$, define the $\mathcal{G}$ -measurable random variable $Z = A/B$. Since conditional expectation is positive,

$$0 \leq \mathbb{E}((X - ZY)^2 \mid \mathcal{G}).$$

Using linearity and the rule for taking out $\mathcal{G}$ -measurable factors,

$$\begin{aligned} 0 &\leq \mathbb{E}(X^2 \mid \mathcal{G}) - 2Z\mathbb{E}(XY \mid \mathcal{G}) + Z^2\mathbb{E}(Y^2 \mid \mathcal{G}) \\ &= C - 2(A/B)A + (A^2/B^2)B \\ &= C - \frac{A^2}{B}. \end{aligned}$$

Multiplying by $B > 0$ gives $A^2 \leq BC$ on $\{B > 0\}$, while the same inequality is already true on $\{B = 0\}$. Hence

$$\mathbb{E}(XY \mid \mathcal{G})^2 \leq \mathbb{E}(X^2 \mid \mathcal{G}) \mathbb{E}(Y^2 \mid \mathcal{G}).$$

Exercise 4.1.4

Question. Using regular conditional probabilities, prove the conditional Holder inequality: if $1 < p, q < \infty$ and $p^{-1} + q^{-1} = 1$, then

$$\mathbb{E}(|XY| \mid \mathcal{G}) \leq \mathbb{E}(|X|^p \mid \mathcal{G})^{1/p} \mathbb{E}(|Y|^q \mid \mathcal{G})^{1/q}.$$

Knowledge used. `durrett_ch4_regular_conditional_distribution`;
`durrett_ch4_conditional_expectation_definition`;
`durrett_ch4_conditional_expectation_properties`.

Solution. Let $\mu(\omega, \cdot)$ be a regular conditional distribution of (X, Y) given $\mathcal{G}$. Then, for almost every ω ,

$$\mathbb{E}(|XY| \mid \mathcal{G})(\omega) = \int_{\mathbb{R}^2} |xy| \mu(\omega, d(x, y)).$$

For those ω , ordinary Holder's inequality applied to the probability measure $\mu(\omega, \cdot)$ gives

$$\int_{\mathbb{R}^2} |xy| \mu(\omega, d(x, y)) \leq \left(\int_{\mathbb{R}^2} |x|^p \mu(\omega, d(x, y)) \right)^{1/p} \left(\int_{\mathbb{R}^2} |y|^q \mu(\omega, d(x, y)) \right)^{1/q}.$$

The regular conditional distribution formula identifies the two factors as $\mathbb{E}(|X|^p \mid \mathcal{G})(\omega)^{1/p}$ and $\mathbb{E}(|Y|^q \mid \mathcal{G})(\omega)^{1/q}$. Therefore the desired inequality holds almost surely.

Exercise 4.1.5

Question. Give an example on $\Omega = \{a, b, c\}$ for which

$$\mathbb{E}(\mathbb{E}(X \mid \mathcal{F}_1) \mid \mathcal{F}_2) \neq \mathbb{E}(\mathbb{E}(X \mid \mathcal{F}_2) \mid \mathcal{F}_1).$$

Knowledge used. `durrett_ch4_partition_conditioning_formula`;
`durrett_ch4_tower_property_smaller_sigma_field_wins`.

Solution. Let $\mathbb{P}$ be uniform on $\Omega = \{a, b, c\}$. Define

$$\mathcal{F}_1 = \sigma(\{a, b\}, \{c\}), \quad \mathcal{F}_2 = \sigma(\{a, c\}, \{b\}),$$

and let $X = \mathbf{1}_{\{a\}}$.

First compute $\mathbb{E}(X \mid \mathcal{F}_1)$. It is constant on the atoms of $\mathcal{F}_1$:

$$\mathbb{E}(X \mid \mathcal{F}_1) = \begin{cases} 1/2, & \omega \in \{a, b\}, \\ 0, & \omega = c. \end{cases}$$

Conditioning this on $\mathcal{F}_2$ gives

$$\mathbb{E}(\mathbb{E}(X \mid \mathcal{F}_1) \mid \mathcal{F}_2) = \begin{cases} 1/4, & \omega \in \{a, c\}, \\ 1/2, & \omega = b. \end{cases}$$

On the other hand,

$$\mathbb{E}(X \mid \mathcal{F}_2) = \begin{cases} 1/2, & \omega \in \{a, c\}, \\ 0, & \omega = b. \end{cases}$$

Conditioning this on $\mathcal{F}_1$ gives

$$\mathbb{E}(\mathbb{E}(X \mid \mathcal{F}_2) \mid \mathcal{F}_1) = \begin{cases} 1/4, & \omega \in \{a, b\}, \\ 1/2, & \omega = c. \end{cases}$$

The two random variables differ, for example at b and c . This does not contradict the tower property, because neither $\mathcal{F}_1 \subseteq \mathcal{F}_2$ nor $\mathcal{F}_2 \subseteq \mathcal{F}_1$.

Exercise 4.1.6

Question. Suppose $\mathcal{G} \subseteq \mathcal{F}$ and $\mathbb{E}X^2 < \infty$. Prove

$$\mathbb{E}\{X - \mathbb{E}(X \mid \mathcal{F})\}^2 + \mathbb{E}\{\mathbb{E}(X \mid \mathcal{F}) - \mathbb{E}(X \mid \mathcal{G})\}^2 = \mathbb{E}\{X - \mathbb{E}(X \mid \mathcal{G})\}^2.$$

Knowledge used. [durrett_ch4_conditional_expectation_l2_projection](#);
[durrett_ch4_taking_out_what_is_known](#); [durrett_ch4_tower_property_smaller_sigma_field_wins](#).

Solution. Let

$$Y = \mathbb{E}(X \mid \mathcal{F}), \quad Z = \mathbb{E}(X \mid \mathcal{G}).$$

Since $\mathcal{G} \subseteq \mathcal{F}$, both Y and Z are $\mathcal{F}$ -measurable. Also $Y - Z \in L^2(\mathcal{F})$. The L^2 projection property of conditional expectation gives

$$\mathbb{E}[(X - Y)W] = 0 \quad \text{for every } W \in L^2(\mathcal{F}).$$

Taking $W = Y - Z$, we get

$$\mathbb{E}[(X - Y)(Y - Z)] = 0.$$

Now write

$$X - Z = (X - Y) + (Y - Z).$$

Squaring and taking expectations gives

$$\begin{aligned} \mathbb{E}(X - Z)^2 &= \mathbb{E}(X - Y)^2 + \mathbb{E}(Y - Z)^2 + 2\mathbb{E}[(X - Y)(Y - Z)] \\ &= \mathbb{E}(X - Y)^2 + \mathbb{E}(Y - Z)^2. \end{aligned}$$

Substituting back $Y = \mathbb{E}(X \mid \mathcal{F})$ and $Z = \mathbb{E}(X \mid \mathcal{G})$ proves the identity.

Exercise 4.1.7

Question. Let

$$\text{Var}(X \mid \mathcal{F}) = \mathbb{E}(X^2 \mid \mathcal{F}) - \mathbb{E}(X \mid \mathcal{F})^2.$$

Prove the variance decomposition

$$\text{Var}(X) = \mathbb{E}[\text{Var}(X \mid \mathcal{F})] + \text{Var}(\mathbb{E}(X \mid \mathcal{F})).$$

Knowledge used. [durrett_ch4_conditional_expectation_properties](#);
[durrett_ch4_conditional_expectation_l2_projection](#).

Solution. Let $Y = \mathbb{E}(X \mid \mathcal{F})$. Taking expectations in the definition of conditional variance gives

$$\mathbb{E}[\text{Var}(X \mid \mathcal{F})] = \mathbb{E}[\mathbb{E}(X^2 \mid \mathcal{F})] - \mathbb{E}(Y^2) = \mathbb{E}X^2 - \mathbb{E}Y^2.$$

Also

$$\text{Var}(Y) = \mathbb{E}Y^2 - (\mathbb{E}Y)^2.$$

Since $\mathbb{E}Y = \mathbb{E}[\mathbb{E}(X \mid \mathcal{F})] = \mathbb{E}X$, adding the two displayed identities gives

$$\mathbb{E}[\text{Var}(X \mid \mathcal{F})] + \text{Var}(Y) = \mathbb{E}X^2 - (\mathbb{E}X)^2 = \text{Var}(X).$$

Exercise 4.1.8

Question. Let $Y_1, Y_2, \dots$ be i.i.d. with mean μ and variance σ^2 . Let N be an independent positive integer-valued random variable with $\mathbb{E}N^2 < \infty$, and define

$$X = Y_1 + \dots + Y_N.$$

Show that

$$\text{Var}(X) = \sigma^2 \mathbb{E}N + \mu^2 \text{Var}(N).$$

Knowledge used. `durrett_ch4_independent_conditioning_rule`;
`durrett_ch4_conditional_expectation_properties`;
`durrett_ch4_tower_property_smaller_sigma_field_wins`;
`durrett_ch4_conditional_expectation_l2_projection`.

Solution. Condition on $\sigma(N)$. Given $N = n$, the sum X has mean $n\mu$ and variance $n\sigma^2$. Thus

$$\mathbb{E}(X \mid N) = \mu N$$

and

$$\text{Var}(X \mid N) = \sigma^2 N.$$

Using the variance decomposition from Exercise 4.1.7,

$$\begin{aligned} \text{Var}(X) &= \mathbb{E}[\text{Var}(X \mid N)] + \text{Var}(\mathbb{E}(X \mid N)) \\ &= \mathbb{E}[\sigma^2 N] + \text{Var}(\mu N) \\ &= \sigma^2 \mathbb{E}N + \mu^2 \text{Var}(N). \end{aligned}$$

Exercise 4.1.9

Question. Suppose $\mathbb{E}(Y \mid \mathcal{G}) = X$ and $\mathbb{E}Y^2 = \mathbb{E}X^2 < \infty$. Prove that $X = Y$ almost surely.

Knowledge used. `durrett_ch4_conditional_expectation_l2_projection`;
`durrett_ch4_taking_out_what_is_known`; `durrett_ch4_conditional_expectation_properties`.

Solution. Since $X = \mathbb{E}(Y \mid \mathcal{G})$, the random variable X is $\mathcal{G}$ -measurable. Using the rule for taking out what is known,

$$\mathbb{E}(XY) = \mathbb{E}[X \mathbb{E}(Y \mid \mathcal{G})] = \mathbb{E}(X^2).$$

Therefore

$$\begin{aligned} \mathbb{E}(Y - X)^2 &= \mathbb{E}Y^2 - 2\mathbb{E}(XY) + \mathbb{E}X^2 \\ &= \mathbb{E}Y^2 - 2\mathbb{E}X^2 + \mathbb{E}X^2 \\ &= \mathbb{E}Y^2 - \mathbb{E}X^2 \\ &= 0. \end{aligned}$$

Hence $Y - X = 0$ almost surely.

Exercise 4.1.10

Question. Assume $\mathbb{E}|Y| < \infty$ and $\mathbb{E}(Y \mid \mathcal{G})$ has the same distribution as Y . Prove that

$$\mathbb{E}(Y \mid \mathcal{G}) = Y \quad \text{a.s.}$$

Knowledge used. [durrett_ch4_conditional_expectation_properties](#);
[durrett_ch4_conditional_jensen_contraction](#); [durrett_ch4_taking_out_what_is_known](#).

Solution. Let $Z = \mathbb{E}(Y \mid \mathcal{G})$. We first prove a sign lemma. Suppose $U \in L^1$, $M = \mathbb{E}(U \mid \mathcal{G})$, and M has the same distribution as U . Then

$$\mathbb{E}|M| = \mathbb{E}|U|.$$

Since $\text{sgn}(M)$ is $\mathcal{G}$ -measurable and bounded,

$$\mathbb{E}|M| = \mathbb{E}[\text{sgn}(M)M] = \mathbb{E}[\text{sgn}(M)\mathbb{E}(U \mid \mathcal{G})] = \mathbb{E}[\text{sgn}(M)U] \leq \mathbb{E}|U|.$$

The first and last quantities are equal, so equality must hold in $\text{sgn}(M)U \leq |U|$. Therefore $\text{sgn}(M)U = |U|$ almost surely. In particular, U and M have the same sign almost surely, with both zero on the set where $M = 0$.

Now apply the lemma to

$$U_c = Y - c, \quad M_c = Z - c = \mathbb{E}(Y - c \mid \mathcal{G}),$$

where $c \in \mathbb{Q}$. Because Z and Y have the same distribution, M_c and U_c also have the same distribution. Hence, for each rational c ,

$$\text{sgn}(Y - c) = \text{sgn}(Z - c) \quad \text{a.s.}$$

Outside one null set, this equality holds for every rational c . If $Y(\omega) < Z(\omega)$, choose a rational c strictly between them; then $Y(\omega) - c < 0$ while $Z(\omega) - c > 0$, a contradiction. The case $Z(\omega) < Y(\omega)$ is the same. Hence $Y = Z$ almost surely, i.e.

$$\mathbb{E}(Y \mid \mathcal{G}) = Y \quad \text{a.s.}$$

Exercise 4.2.1

Question. Suppose (X_n) is a martingale with respect to a filtration $(\mathcal{G}_n)$. Let

$$\mathcal{F}_n = \sigma(X_1, \dots, X_n).$$

Show that $\mathcal{G}_n \supseteq \mathcal{F}_n$ and that (X_n) is also a martingale with respect to $(\mathcal{F}_n)$.

Knowledge used. [durrett_ch4_martingale_definition](#);
[durrett_ch4_tower_property_smaller_sigma_field_wins](#);
[durrett_ch4_multi_step_martingale_property](#).

Solution. Since X_i is $\mathcal{G}_i$ -measurable and $\mathcal{G}_i \subseteq \mathcal{G}_n$ for $i \leq n$, each X_i with $i \leq n$ is $\mathcal{G}_n$ -measurable. Hence

$$\mathcal{F}_n = \sigma(X_1, \dots, X_n) \subseteq \mathcal{G}_n.$$

The process is integrable by hypothesis, and X_n is $\mathcal{F}_n$ -measurable by definition. Finally, using the tower property and $\mathcal{F}_n \subseteq \mathcal{G}_n$,

$$\mathbb{E}(X_{n+1} \mid \mathcal{F}_n) = \mathbb{E}\{\mathbb{E}(X_{n+1} \mid \mathcal{G}_n) \mid \mathcal{F}_n\} = \mathbb{E}(X_n \mid \mathcal{F}_n) = X_n.$$

Thus (X_n) is a martingale with respect to $(\mathcal{F}_n)$.

Exercise 4.2.2

Question. Give an example of a submartingale (X_n) such that (X_n^2) is a supermartingale.

Knowledge used. [durrett_ch4_martingale_definition](#);
[durrett_ch4_conditional_expectation_properties](#).

Solution. Take a deterministic process

$$X_n = -\frac{1}{n+1}, \quad n \geq 0,$$

on any probability space with the trivial filtration. Since $X_{n+1} \geq X_n$, we have

$$\mathbb{E}(X_{n+1} \mid \mathcal{F}_n) = X_{n+1} \geq X_n,$$

so (X_n) is a submartingale. But

$$X_{n+1}^2 = \frac{1}{(n+2)^2} \leq \frac{1}{(n+1)^2} = X_n^2,$$

so

$$\mathbb{E}(X_{n+1}^2 \mid \mathcal{F}_n) = X_{n+1}^2 \leq X_n^2.$$

Therefore (X_n^2) is a supermartingale.

Exercise 4.2.3

Question. Show that if (X_n) and (Y_n) are submartingales with respect to $(\mathcal{F}_n)$, then $(X_n \vee Y_n)$ is also a submartingale.

Knowledge used. [durrett_ch4_martingale_definition](#);
[durrett_ch4_conditional_expectation_properties](#);
[durrett_ch4_convex_transform_submartingale](#).

Solution. The random variable $X_n \vee Y_n$ is $\mathcal{F}_n$ -measurable and integrable because

$$|X_n \vee Y_n| \leq |X_n| + |Y_n|.$$

Also,

$$X_{n+1} \vee Y_{n+1} \geq X_{n+1}, \quad X_{n+1} \vee Y_{n+1} \geq Y_{n+1}.$$

Taking conditional expectations and using monotonicity,

$$\mathbb{E}(X_{n+1} \vee Y_{n+1} \mid \mathcal{F}_n) \geq \mathbb{E}(X_{n+1} \mid \mathcal{F}_n) \geq X_n,$$

and similarly

$$\mathbb{E}(X_{n+1} \vee Y_{n+1} \mid \mathcal{F}_n) \geq Y_n.$$

Hence

$$\mathbb{E}(X_{n+1} \vee Y_{n+1} \mid \mathcal{F}_n) \geq X_n \vee Y_n,$$

so $(X_n \vee Y_n)$ is a submartingale.

Exercise 4.2.4

Question. Let $(X_n)_{n \geq 0}$ be a submartingale with $\sup_n X_n < \infty$ almost surely. Put

$$\xi_n = X_n - X_{n-1}, \quad n \geq 1,$$

and suppose

$$\mathbb{E} \left(\sup_{n \geq 1} \xi_n^+ \right) < \infty.$$

Show that X_n converges almost surely.

Knowledge used. [durrett_ch4_stopped_process_supermartingale](#);
[durrett_ch4_upcrossing_inequality](#); [durrett_ch4_martingale_convergence_theorem](#).

Solution. Let

$$M = \sup_{n \geq 1} \xi_n^+.$$

Then $M \in L^1$ by assumption. Fix a real number c and define the stopping time

$$T_c = \inf\{n \geq 0 : X_n \geq c\}.$$

The stopped process $X_{n \wedge T_c}$ is a submartingale. Moreover, on the event $\{T_c < \infty\}$, if $T_c \geq 1$, then

$$X_{T_c} = X_{T_c-1} + \xi_{T_c} < c + \xi_{T_c}^+ \leq c + M,$$

while if $T_c = 0$ the stopped value is fixed at X_0 . Thus the positive parts of the stopped process are bounded by an integrable random variable depending on c , X_0 , and M . Hence

$$\sup_n \mathbb{E}(X_{n \wedge T_c}^+) < \infty.$$

By the martingale convergence theorem, the stopped process $X_{n \wedge T_c}$ converges almost surely. Equivalently, for every rational interval $[a, b]$, the stopped process has only finitely many upcrossings of $[a, b]$ almost surely.

Since $\sup_n X_n < \infty$ almost surely, the events

$$A_c = \{\sup_n X_n < c\}, \quad c \in \mathbb{N},$$

increase to an event of probability one. On A_c , the stopped process at T_c is just the original process, so the original process has only finitely many upcrossings of every rational interval $[a, b]$. Therefore

$$\liminf_{n \rightarrow \infty} X_n = \limsup_{n \rightarrow \infty} X_n$$

almost surely. Thus X_n converges almost surely.

Exercise 4.2.5

Question. Give an example of a martingale (X_n) such that

$$X_n \rightarrow -\infty \quad \text{a.s.}$$

Knowledge used. [durrett_ch4_martingale_definition](#); [durrett_ch4_random_walk_martingales](#).

Solution. Let $\xi_1, \xi_2, \dots$ be independent random variables with

$$\mathbb{P}(\xi_n = -1) = 1 - \frac{1}{n^2}, \quad \mathbb{P}(\xi_n = n^2 - 1) = \frac{1}{n^2}.$$

Then

$$\mathbb{E}\xi_n = -\left(1 - \frac{1}{n^2}\right) + (n^2 - 1)\frac{1}{n^2} = 0.$$

Define

$$X_n = \sum_{k=1}^n \xi_k, \quad \mathcal{F}_n = \sigma(\xi_1, \dots, \xi_n).$$

Then (X_n) is a martingale.

Since

$$\sum_{n=1}^{\infty} \mathbb{P}(\xi_n \neq -1) = \sum_{n=1}^{\infty} \frac{1}{n^2} < \infty,$$

the Borel-Cantelli lemma implies that only finitely many of the large positive jumps occur almost surely. After the last such jump, all increments are equal to -1 , so $X_n \rightarrow -\infty$ almost surely.

Exercise 4.2.6

Question. Let $Y_1, Y_2, \dots$ be nonnegative i.i.d. random variables with $\mathbb{E}Y_m = 1$ and $\mathbb{P}(Y_m = 1) < 1$. Example 4.2.3 shows that

$$X_n = \prod_{m=1}^n Y_m$$

is a martingale.

- (i) Use the nonnegative supermartingale convergence theorem and a contradiction argument to show that $X_n \rightarrow 0$ almost surely.
- (ii) Use the strong law of large numbers to identify the exponential rate of decay.

Knowledge used. [durrett_ch4_random_walk_martingales](#);
[durrett_ch4_nonnegative_supermartingale_convergence](#);
[durrett_ch4_martingale_convergence_theorem](#).

Solution. Since (X_n) is a nonnegative martingale, it is also a nonnegative supermartingale. Therefore $X_n \rightarrow X_\infty$ almost surely for some finite random variable $X_\infty \geq 0$.

We claim that $X_\infty = 0$ a.s. Since Y_1 is not almost surely equal to 1, there is an $\varepsilon > 0$ such that

$$p = \mathbb{P}(|Y_1 - 1| > \varepsilon) > 0.$$

By independence and the second Borel-Cantelli lemma, $|Y_n - 1| > \varepsilon$ occurs infinitely often almost surely. If $X_\infty > 0$, then eventually $X_n > 0$ and

$$Y_n = \frac{X_n}{X_{n-1}} \rightarrow 1,$$

because both X_n and X_{n-1} converge to the same positive limit. This contradicts the infinitely many deviations of Y_n from 1. Thus $X_\infty = 0$ a.s.

For the rate, suppose first that $\log Y_1$ is integrable. Then

$$\frac{1}{n} \log X_n = \frac{1}{n} \sum_{m=1}^n \log Y_m \rightarrow c := \mathbb{E} \log Y_1 \quad \text{a.s.}$$

By Jensen's inequality,

$$c = \mathbb{E} \log Y_1 < \log \mathbb{E} Y_1 = 0,$$

where the inequality is strict because Y_1 is not almost surely constant. If $\mathbb{P}(Y_1 = 0) > 0$, then $X_n = 0$ eventually almost surely, and the same conclusion should be read as decay with rate $-\infty$.

Exercise 4.2.7

Question. Suppose $y_n > -1$ for all n and

$$\sum_{n=1}^{\infty} |y_n| < \infty.$$

Show that the infinite product

$$\prod_{n=1}^{\infty} (1 + y_n)$$

exists.

Knowledge used. [durrett_ch4_nonnegative_supermartingale_convergence](#).

Solution. Since $\sum_n |y_n| < \infty$, we have $y_n \rightarrow 0$. Hence, for all large n , $|y_n| \leq 1/2$. For such n ,

$$|\log(1 + y_n)| \leq 2|y_n|.$$

Thus the series

$$\sum_{n=1}^{\infty} \log(1 + y_n)$$

converges absolutely after deleting finitely many initial terms. Since each factor $1 + y_n$ is positive, the partial products satisfy

$$\prod_{n=1}^N (1 + y_n) = \exp \left\{ \sum_{n=1}^N \log(1 + y_n) \right\}.$$

The exponent converges to a finite limit, so the product converges to a finite positive limit.

Exercise 4.2.8

Question. Let (X_n) and (Y_n) be positive integrable processes adapted to $(\mathcal{F}_n)$. Suppose

$$\mathbb{E}(X_{n+1} \mid \mathcal{F}_n) \leq (1 + Y_n)X_n$$

and

$$\sum_{n=0}^{\infty} Y_n < \infty \quad \text{a.s.}$$

Prove that X_n converges almost surely to a finite limit by finding a related supermartingale.

Knowledge used. [durrett_ch4_taking_out_what_is_known](#);
[durrett_ch4_nonnegative_supermartingale_convergence](#); [durrett_ch4_martingale_definition](#).

Solution. Define

$$A_0 = 1, \quad A_n = \prod_{m=0}^{n-1} (1 + Y_m), \quad n \geq 1,$$

and put

$$Z_n = \frac{X_n}{A_n}.$$

Since A_n is $\mathcal{F}_n$ -measurable and positive,

$$\begin{aligned} \mathbb{E}(Z_{n+1} \mid \mathcal{F}_n) &= \mathbb{E}\left(\frac{X_{n+1}}{A_n(1 + Y_n)} \mid \mathcal{F}_n\right) \\ &= \frac{1}{A_n(1 + Y_n)} \mathbb{E}(X_{n+1} \mid \mathcal{F}_n) \\ &\leq \frac{X_n}{A_n} = Z_n. \end{aligned}$$

Thus (Z_n) is a positive supermartingale. By the nonnegative supermartingale convergence theorem, $Z_n \rightarrow Z_\infty$ almost surely for some finite Z_∞ .

By Exercise 4.2.7, the assumption $\sum_n Y_n < \infty$ implies $A_n \rightarrow A_\infty$, where $0 < A_\infty < \infty$ almost surely. Therefore

$$X_n = A_n Z_n \rightarrow A_\infty Z_\infty$$

almost surely, and the limit is finite.

Exercise 4.2.9

Question. Prove the switching principle. Suppose (X_n^1) and (X_n^2) are supermartingales with respect to $(\mathcal{F}_n)$, and N is a stopping time such that

$$X_N^1 \geq X_N^2.$$

Show that both

$$Y_n = X_n^1 \mathbf{1}_{\{N > n\}} + X_n^2 \mathbf{1}_{\{N \leq n\}}$$

and

$$Z_n = X_n^1 \mathbf{1}_{\{N \geq n\}} + X_n^2 \mathbf{1}_{\{N < n\}}$$

are supermartingales.

Knowledge used. [durrett_ch4_martingale_definition](#);
[durrett_ch4_taking_out_what_is_known](#); [durrett_ch4_stopped_process_supermartingale](#).

Solution. Both processes are adapted and integrable. We first prove the claim for Y . On $\{N \leq n\}$,

$$Y_n = X_n^2, \quad \mathbb{E}(Y_{n+1} \mid \mathcal{F}_n) \leq \mathbb{E}(X_{n+1}^2 \mid \mathcal{F}_n) \leq X_n^2 = Y_n.$$

On $\{N > n\}$, either $N > n + 1$, in which case $Y_{n+1} = X_{n+1}^1$, or $N = n + 1$, in which case

$$Y_{n+1} = X_{n+1}^2 \leq X_{n+1}^1$$

by the switching assumption at time N . Hence on $\{N > n\}$, $Y_{n+1} \leq X_{n+1}^1$, and so

$$\mathbb{E}(Y_{n+1} \mid \mathcal{F}_n) \leq \mathbb{E}(X_{n+1}^1 \mid \mathcal{F}_n) \leq X_n^1 = Y_n.$$

Combining the two $\mathcal{F}_n$ -measurable cases shows that Y is a supermartingale.

For Z , split into the events $\{N < n\}$, $\{N = n\}$, and $\{N > n\}$. On $\{N < n\}$, the process follows X^2 , so the supermartingale inequality is immediate. On $\{N > n\}$, the process follows X^1 at time $n+1$, so the inequality is also immediate. On $\{N = n\}$,

$$Z_n = X_n^1, \quad Z_{n+1} = X_{n+1}^2,$$

and therefore

$$\mathbb{E}(Z_{n+1} \mid \mathcal{F}_n) = \mathbb{E}(X_{n+1}^2 \mid \mathcal{F}_n) \leq X_n^2 \leq X_n^1 = Z_n.$$

Thus Z is a supermartingale.

Exercise 4.2.10

Question. Prove Dubins' inequality. For every positive supermartingale $(X_n)_{n \geq 0}$, if U is the number of upcrossings of $[a, b]$ with $0 < a < b$, then

$$\mathbb{P}(U \geq k) \leq \left(\frac{a}{b}\right)^k \mathbb{E} \min(X_0/a, 1).$$

Knowledge used. `durrett_ch4_nonnegative_supermartingale_convergence`;
`durrett_ch4_upcrossing_inequality`; `durrett_ch4_martingale_transform_no_gambling_system`;
`durrett_ch4_stopped_process_supermartingale`.

Solution. Let

$$N_0 = -1,$$

and for $j \geq 1$ define the stopping times

$$N_{2j-1} = \inf\{m > N_{2j-2} : X_m \leq a\}, \quad N_{2j} = \inf\{m > N_{2j-1} : X_m \geq b\}.$$

Define Y_n by

$$Y_n = 1, \quad 0 \leq n < N_1,$$

and, for $j \geq 1$,

$$Y_n = \begin{cases} (b/a)^{j-1} X_n/a, & N_{2j-1} \leq n < N_{2j}, \\ (b/a)^j, & N_{2j} \leq n < N_{2j+1}. \end{cases}$$

At a time N_{2j-1} , we switch from the constant value $(b/a)^{j-1}$ to $(b/a)^{j-1} X_n/a$, and this switch is downward because $X_{N_{2j-1}} \leq a$. At a time N_{2j} , we switch from $(b/a)^{j-1} X_n/a$ to the constant $(b/a)^j$, and this switch is also downward because $X_{N_{2j}} \geq b$. Since positive constants and positive multiples of the supermartingale X are supermartingales, the switching principle implies inductively that

$$Z_n^j = Y_{n \wedge N_j}$$

is a supermartingale for each fixed j .

In particular, for $j = 2k$,

$$\mathbb{E}Y_{n \wedge N_{2k}} \leq \mathbb{E}Y_0.$$

From the definition,

$$Y_0 = \min(X_0/a, 1).$$

Letting $n \rightarrow \infty$ and using Fatou's lemma,

$$\mathbb{E} \left[\liminf_{n \rightarrow \infty} Y_{n \wedge N_{2k}} \right] \leq \mathbb{E} \min(X_0/a, 1).$$

On the event $\{U \geq k\}$, the stopping time N_{2k} is finite, and

$$\liminf_{n \rightarrow \infty} Y_{n \wedge N_{2k}} = Y_{N_{2k}} \geq (b/a)^k.$$

Therefore

$$(b/a)^k \mathbb{P}(U \geq k) \leq \mathbb{E} \min(X_0/a, 1),$$

which is equivalent to

$$\mathbb{P}(U \geq k) \leq \left(\frac{a}{b}\right)^k \mathbb{E} \min(X_0/a, 1).$$

Exercise 4.3.1

Question. Give a bounded process with bounded increments that visits each of $-1, 0, 1$ infinitely often almost surely. Explain why this shows that bounded increments alone are not enough in Theorem 4.3.1.

Knowledge used. [durrett_ch4_bounded_increment_convergence](#);
[durrett_ch4_martingale_convergence_theorem](#).

Solution. Take the deterministic sequence

$$X_{3k} = -1, \quad X_{3k+1} = 0, \quad X_{3k+2} = 1, \quad k \geq 0.$$

Then $\sup_n |X_n| \leq 1$ and $\sup_n |X_{n+1} - X_n| \leq 2$. Each of $-1, 0, 1$ occurs infinitely often, so

$$\mathbb{P}(X_n = a \text{ i.o.}) = 1, \quad a \in \{-1, 0, 1\}.$$

The sequence does not converge and does not oscillate between $+\infty$ and $-\infty$. Thus the martingale hypothesis in Theorem 4.3.1 is essential; bounded increments by themselves do not force the theorem's dichotomy. A literal bounded martingale example with these properties cannot exist, since bounded martingales converge almost surely.

Exercise 4.3.2

Question. Construct a finite-state Markov chain on $\{-1, 0, 1\}$ such that

$$\mathbb{P}(X_n = 0) \rightarrow 1 - 2p, \quad \mathbb{P}(X_n = -1) \rightarrow p, \quad \mathbb{P}(X_n = 1) \rightarrow p$$

for some chosen $p \in (0, 1/2)$, while the chain does not converge almost surely.

Knowledge used. [durrett_ch4_bounded_increment_convergence](#);
[durrett_ch4_martingale_convergence_theorem](#).

Solution. Choose, for instance, $p = 1/4$. Let (X_n) be independent with

$$\mathbb{P}(X_n = -1) = 1/4, \quad \mathbb{P}(X_n = 0) = 1/2, \quad \mathbb{P}(X_n = 1) = 1/4$$

for every n . This is a Markov chain whose transition probabilities do not depend on the present state: every row of the transition matrix is $(1/4, 1/2, 1/4)$.

The distribution is constant in n , so it converges to the desired limits with $p = 1/4$. Since the events $\{X_n = a\}$ are independent and have positive probability for each $a \in \{-1, 0, 1\}$, the second Borel-Cantelli lemma implies that each state is visited infinitely often almost surely. Hence X_n does not converge almost surely, and in fact does not converge in probability. This illustrates that convergence in distribution alone does not imply almost sure convergence.

Exercise 4.3.3

Question. Let (X_n) and (Y_n) be positive integrable adapted processes. Suppose

$$\mathbb{E}(X_{n+1} \mid \mathcal{F}_n) \leq X_n + Y_n$$

and

$$\sum_{n=1}^{\infty} Y_n < \infty \quad \text{a.s.}$$

Prove that X_n converges almost surely to a finite limit.

Knowledge used. [durrett_ch4_martingale_definition](#);
[durrett_ch4_stopped_process_supermartingale](#);
[durrett_ch4_nonnegative_supermartingale_convergence](#).

Solution. Let

$$S_n = \sum_{m=1}^n Y_m, \quad Z_n = X_n - S_n.$$

This indexing puts Y_{n+1} outside $\mathcal{F}_n$, so use the predictable partial sum instead:

$$T_n = \sum_{m=0}^{n-1} Y_m, \quad Z_n = X_n - T_n,$$

where the hypothesis is read as $\mathbb{E}(X_{n+1} \mid \mathcal{F}_n) \leq X_n + Y_n$. Then

$$\mathbb{E}(Z_{n+1} \mid \mathcal{F}_n) = \mathbb{E}(X_{n+1} \mid \mathcal{F}_n) - T_n - Y_n \leq X_n - T_n = Z_n.$$

Thus Z_n is a supermartingale.

Fix $M < \infty$ and stop just before the partial sum of the Y 's exceeds M :

$$N_M = \inf\{n : T_{n+1} > M\}.$$

For $n \leq N_M$, $T_n \leq M$, and since $X_n \geq 0$,

$$Z_{n \wedge N_M} = X_{n \wedge N_M} - T_{n \wedge N_M} \geq -M.$$

Hence $Z_{n \wedge N_M} + M$ is a nonnegative supermartingale, so it converges almost surely. On the event $\{T_\infty \leq M\}$, this stopped process is eventually the original Z_n , so Z_n converges there. Since $T_\infty < \infty$ almost surely, the events $\{T_\infty \leq M\}$ increase to probability one. Therefore Z_n converges almost surely. Also $T_n \rightarrow T_\infty < \infty$, so

$$X_n = Z_n + T_n$$

converges almost surely to a finite limit.

Exercise 4.3.4

Question. Let $p_m \in [0, 1)$. Show that

$$\prod_{m=1}^{\infty} (1 - p_m) = 0 \quad \text{if and only if} \quad \sum_{m=1}^{\infty} p_m = \infty.$$

Knowledge used. [durrett_ch4_conditional_borel_cantelli](#);
[durrett_ch4_nonnegative_supermartingale_convergence](#).

Solution. If $\sum_m p_m < \infty$, then $p_m \rightarrow 0$. For all large m , $p_m \leq 1/2$, and

$$\log(1 - p_m) \geq -2p_m.$$

Thus $\sum_m \log(1 - p_m)$ converges to a finite real number after discarding finitely many initial terms. Since each initial factor is strictly positive, the product is strictly positive.

Conversely, if $\sum_m p_m = \infty$, let A_m be independent events with $\mathbb{P}(A_m) = p_m$. By the second Borel-Cantelli lemma, $\mathbb{P}(A_m \text{ i.o.}) = 1$. In particular, the probability that none of the events ever occurs is zero. But

$$\mathbb{P}(A_m^c \text{ for all } m) = \prod_{m=1}^{\infty} (1 - p_m).$$

Hence the product is zero.

Exercise 4.3.5

Question. Show that if

$$\sum_{n=2}^{\infty} \mathbb{P}\left(A_n \mid \bigcap_{m=1}^{n-1} A_m^c\right) = \infty,$$

then

$$\mathbb{P}\left(\bigcap_{m=1}^{\infty} A_m^c\right) = 0.$$

Knowledge used. [durrett_ch4_conditional_borel_cantelli](#);
[durrett_ch4_conditional_expectation_definition](#).

Solution. Let

$$B_n = \bigcap_{m=1}^{n-1} A_m^c, \quad q_n = \mathbb{P}(A_n \mid B_n),$$

whenever $\mathbb{P}(B_n) > 0$. If $\mathbb{P}(B_n) = 0$ for some n , then the conclusion is immediate. Otherwise,

$$\mathbb{P}(B_{n+1}) = \mathbb{P}(B_n) \mathbb{P}(A_n^c \mid B_n) = \mathbb{P}(B_n)(1 - q_n).$$

Iterating gives

$$\mathbb{P}\left(\bigcap_{m=1}^N A_m^c\right) = \mathbb{P}(A_1^c) \prod_{n=2}^N (1 - q_n).$$

By Exercise 4.3.4, the infinite product is zero because $\sum_n q_n = \infty$. Letting $N \rightarrow \infty$ gives

$$\mathbb{P}\left(\bigcap_{m=1}^{\infty} A_m^c\right) = 0.$$

Exercise 4.3.6

Question. For the partition approximation in Example 4.3.7, verify directly that X_n is a martingale. If the condition $\mu_n \ll \nu_n$ is dropped and X_n is set to 0 on atoms with ν -mass zero, show that X_n is a supermartingale.

Knowledge used. [durrett_ch4_radon_nikodym_derivative_martingale](#);
[durrett_ch4_conditional_expectation_definition](#);
[durrett_ch4_partition_conditioning_formula](#).

Solution. Let $\mathcal{F}_n$ be generated by a finite or countable partition $\{I_{k,n}\}_k$, with the $(n+1)$ st partition refining the n th. In the absolutely continuous case,

$$X_n = \frac{\mu(I_{k,n})}{\nu(I_{k,n})} \quad \text{on } I_{k,n}.$$

Fix an atom $I = I_{k,n}$ with $\nu(I) > 0$. Since the next partition refines the present one,

$$\int_I X_{n+1} d\nu = \sum_{J \subseteq I} \int_J X_{n+1} d\nu = \sum_{J \subseteq I} \mu(J) = \mu(I) = \int_I X_n d\nu.$$

This verifies the conditional expectation identity on every atom of $\mathcal{F}_n$, so $\mathbb{E}_\nu(X_{n+1} \mid \mathcal{F}_n) = X_n$.

If $\mu_n \ll \nu_n$ is not assumed and we define $X_n = 0$ on atoms with ν -mass zero, then for an atom I with $\nu(I) > 0$,

$$\int_I X_{n+1} d\nu = \sum_{\substack{J \subseteq I \\ \nu(J) > 0}} \mu(J) \leq \mu(I) = \int_I X_n d\nu.$$

On atoms with $\nu(I) = 0$, both integrals with respect to ν are zero. Thus

$$\mathbb{E}_\nu(X_{n+1} \mid \mathcal{F}_n) \leq X_n,$$

so (X_n) is a supermartingale.

Exercise 4.3.7

Question. Use Theorem 4.3.5 and the partition construction of Example 4.3.7 to prove the Radon-Nikodym theorem when $\mathcal{F}$ is countably generated.

Knowledge used. [durrett_ch4_radon_nikodym_derivative_martingale](#);
[durrett_ch4_nonnegative_supermartingale_convergence](#);
[durrett_ch4_conditional_expectation_definition](#).

Solution. First suppose μ and ν are finite and $\mu \ll \nu$. Let $\mathcal{F} = \sigma(A_1, A_2, \dots)$ and set

$$\mathcal{F}_n = \sigma(A_1, \dots, A_n).$$

Each $\mathcal{F}_n$ has finitely many atoms. On each atom I of $\mathcal{F}_n$, define

$$X_n = \frac{\mu(I)}{\nu(I)},$$

with the value irrelevant when $\nu(I) = 0$ because $\mu \ll \nu$. By Exercise 4.3.6, (X_n) is the Radon-Nikodym derivative martingale for the finite sigma-fields. Theorem 4.3.5 gives a limit g such that

$$\mu(A) = \int_A g d\nu + \mu(A \cap \{g = \infty\}).$$

But $\nu(g = \infty) = 0$, and $\mu \ll \nu$, so $\mu(\{g = \infty\}) = 0$. Hence

$$\mu(A) = \int_A g \, d\nu, \quad A \in \mathcal{F}.$$

This is the Radon-Nikodym derivative.

For sigma-finite μ and ν , decompose Ω into disjoint measurable sets E_j on which both measures are finite. Apply the finite result to the restrictions of μ and ν to each E_j , obtaining densities g_j on E_j . Then

$$g = \sum_j g_j \mathbf{1}_{E_j}$$

satisfies $\mu(A) = \int_A g \, d\nu$ for all $A \in \mathcal{F}$.

Exercise 4.3.8

Question. For Bernoulli product measures with coordinate probabilities

$$\mu(\xi_n = 1) = \alpha_n, \quad \nu(\xi_n = 1) = \beta_n,$$

find the absolute-continuity criterion from Theorem 4.3.8. Then show that if $0 < \varepsilon \leq \alpha_n, \beta_n \leq 1 - \varepsilon < 1$, the criterion is equivalent to

$$\sum_n (\alpha_n - \beta_n)^2 < \infty.$$

Knowledge used. [durrett_ch4_radon_nikodym_derivative_martingale](#); [durrett_ch4_conditional_jensen_contraction](#).

Solution. For one coordinate, the Hellinger integral in Theorem 4.3.8 is

$$h_n = \sqrt{\alpha_n \beta_n} + \sqrt{(1 - \alpha_n)(1 - \beta_n)}.$$

Thus, assuming the one-coordinate absolute-continuity condition ($\beta_n = 0$ implies $\alpha_n = 0$, and $\beta_n = 1$ implies $\alpha_n = 1$), Theorem 4.3.8 says

$$\mu \ll \nu \quad \Longleftrightarrow \quad \prod_{n=1}^{\infty} h_n > 0.$$

Now assume the uniform bounds away from 0 and 1. Since

$$1 - h_n = \frac{1}{2} \left[(\sqrt{\alpha_n} - \sqrt{\beta_n})^2 + (\sqrt{1 - \alpha_n} - \sqrt{1 - \beta_n})^2 \right],$$

and the functions $x \mapsto \sqrt{x}$ and $x \mapsto \sqrt{1 - x}$ have derivatives bounded above and below away from 0 on $[\varepsilon, 1 - \varepsilon]$, there are constants $0 < c < C < \infty$ such that

$$c(\alpha_n - \beta_n)^2 \leq 1 - h_n \leq C(\alpha_n - \beta_n)^2.$$

Since $0 < h_n \leq 1$, the product $\prod_n h_n$ is positive if and only if $\sum_n (1 - h_n) < \infty$. Therefore, under the uniform bounds,

$$\mu \ll \nu \quad \Longleftrightarrow \quad \sum_n (\alpha_n - \beta_n)^2 < \infty.$$

Exercise 4.3.9

Question. In the Bernoulli product setting, show that if

$$\sum_n \alpha_n < \infty \quad \text{and} \quad \sum_n \beta_n = \infty,$$

then $\mu \perp \nu$. Explain why this shows that the square-summability condition in Exercise 4.3.8 is not sufficient without uniform bounds.

Knowledge used. [durrett_ch4_conditional_borel_cantelli](#);
[durrett_ch4_radon_nikodym_derivative_martingale](#).

Solution. Let

$$A = \{\xi_n = 1 \text{ infinitely often}\}.$$

Under μ , the events $\{\xi_n = 1\}$ are independent and have probabilities α_n . Since $\sum_n \alpha_n < \infty$, the first Borel-Cantelli lemma gives $\mu(A) = 0$.

Under ν , the same events have probabilities β_n , and $\sum_n \beta_n = \infty$. By the second Borel-Cantelli lemma, $\nu(A) = 1$. Hence μ and ν are concentrated on disjoint measurable sets, so $\mu \perp \nu$.

For example, $\alpha_n = n^{-2}$ and $\beta_n = n^{-1}$ satisfy

$$\sum_n (\alpha_n - \beta_n)^2 < \infty,$$

but the preceding argument shows that the product measures are singular.

Exercise 4.3.10

Question. Assume $0 < \alpha_n, \beta_n < 1$ for all n . Show that

$$\sum_n |\alpha_n - \beta_n| < \infty$$

is sufficient for $\mu \ll \nu$.

Knowledge used. [durrett_ch4_radon_nikodym_derivative_martingale](#);
[durrett_ch4_conditional_jensen_contraction](#).

Solution. Since $0 < \alpha_n, \beta_n < 1$, each one-coordinate Bernoulli law for μ is absolutely continuous with respect to the corresponding one for ν . Let

$$h_n = \sqrt{\alpha_n \beta_n} + \sqrt{(1 - \alpha_n)(1 - \beta_n)}.$$

For Bernoulli measures,

$$1 - h_n = \frac{1}{2} \left[(\sqrt{\alpha_n} - \sqrt{\beta_n})^2 + (\sqrt{1 - \alpha_n} - \sqrt{1 - \beta_n})^2 \right].$$

The Hellinger distance is bounded by total variation, and here total variation is $|\alpha_n - \beta_n|$. Thus

$$1 - h_n \leq |\alpha_n - \beta_n|.$$

Therefore $\sum_n (1 - h_n) < \infty$. Since $0 < h_n \leq 1$, this implies

$$\prod_n h_n > 0.$$

Theorem 4.3.8 now gives $\mu \ll \nu$.

Exercise 4.3.11

Question. Let Z_n be a supercritical branching process with mean $\mu > 1$ and let

$$W = \lim_{n \rightarrow \infty} Z_n / \mu^n.$$

Show that if $\mathbb{P}(W = 0) < 1$, then $\mathbb{P}(W = 0) = \rho$, the extinction probability. Conclude that

$$\{W > 0\} = \{Z_n > 0 \text{ for all } n\} \quad \text{a.s.}$$

Knowledge used. [durrett_ch4_branching_process_normalized_martingale](#);
[durrett_ch4_nonnegative_supermartingale_convergence](#).

Solution. Let

$$q = \mathbb{P}(W = 0 \mid Z_0 = 1).$$

Condition on the first generation. If $Z_1 = k$, then the future process is the sum of k independent descendant branching processes. The limit W is zero exactly when all k descendant limits are zero. Hence

$$q = \sum_{k=0}^{\infty} p_k q^k = \phi(q).$$

Thus q is a fixed point of the generating function. In the supercritical case, the fixed points in $[0, 1]$ are the extinction probability ρ and 1. Since $\mathbb{P}(W = 0) < 1$, we have $q < 1$, and therefore $q = \rho$.

Extinction implies $W = 0$, so

$$\{W > 0\} \subseteq \{Z_n > 0 \text{ for all } n\}.$$

Both events have probability $1 - \rho$, so they are equal almost surely.

Exercise 4.3.12

Question. Let (Z_n) be a branching process with offspring generating function

$$\phi(\theta) = \sum_{k=0}^{\infty} p_k \theta^k.$$

Suppose $\rho < 1$ and $\phi(\rho) = \rho$. Show that ρ^{Z_n} is a martingale and conclude that, starting from $Z_0 = x$,

$$\mathbb{P}(Z_n = 0 \text{ for some } n \geq 1) = \rho^x.$$

Knowledge used. [durrett_ch4_branching_process_normalized_martingale](#);
[durrett_ch4_nonnegative_supermartingale_convergence](#);
[durrett_ch4_tower_property_smaller_sigma_field_wins](#).

Solution. Given $\mathcal{F}_n$ and $Z_n = k$, the next generation is the sum of k independent offspring variables. Therefore

$$\mathbb{E}(\rho^{Z_{n+1}} \mid \mathcal{F}_n) = \phi(\rho)^{Z_n} = \rho^{Z_n}.$$

Thus (ρ^{Z_n}) is a bounded martingale.

Let

$$T = \inf\{n \geq 1 : Z_n = 0\}.$$

Since the martingale is bounded, $\mathbb{E}_x \rho^{Z_n} = \rho^x$ for every n . Also $\rho^{Z_n} \rightarrow \mathbf{1}_{\{T < \infty\}}$: on extinction it is eventually 1, while on survival the supercritical branching process has $Z_n \rightarrow \infty$ and hence $\rho^{Z_n} \rightarrow 0$. Bounded convergence gives

$$\rho^x = \lim_{n \rightarrow \infty} \mathbb{E}_x \rho^{Z_n} = \mathbb{P}_x(T < \infty).$$

Exercise 4.3.13

Question. In the Galton-Watson family-name model, each family has exactly three children and each child is male with probability $1/2$. Thus

$$p_0 = 1/8, \quad p_1 = 3/8, \quad p_2 = 3/8, \quad p_3 = 1/8.$$

Compute the extinction probability when $Z_0 = 1$.

Knowledge used. [durrett_ch4_branching_process_normalized_martingale](#).

Solution. The offspring generating function is

$$\phi(s) = \frac{1}{8} + \frac{3}{8}s + \frac{3}{8}s^2 + \frac{1}{8}s^3 = \left(\frac{1+s}{2}\right)^3.$$

The extinction probability ρ is the smallest solution in $[0, 1]$ of

$$\rho = \phi(\rho) = \left(\frac{1+\rho}{2}\right)^3.$$

Equivalently,

$$8\rho = (1+\rho)^3,$$

so

$$\rho^3 + 3\rho^2 - 5\rho + 1 = 0.$$

Factoring,

$$\rho^3 + 3\rho^2 - 5\rho + 1 = (\rho - 1)(\rho^2 + 4\rho - 1).$$

The root in $[0, 1)$ is

$$\rho = -2 + \sqrt{5}.$$

Therefore the probability that the family name dies out, starting from one male, is

$$\boxed{\sqrt{5} - 2}.$$

Exercise 4.4.1

Question. Let (X_n) be a submartingale and let N be a stopping time with $\mathbb{P}(N \leq k) = 1$. Show that if $j \leq k$, then

$$\mathbb{E}(X_j; N = j) \leq \mathbb{E}(X_k; N = j),$$

and sum over j to obtain another proof of $\mathbb{E}X_N \leq \mathbb{E}X_k$.

Knowledge used. [durrett_ch4_bounded_stopping_submartingale_expectation](#);
[durrett_ch4_multi_step_martingale_property](#);
[durrett_ch4_tower_property_smaller_sigma_field_wins](#).

Solution. Since N is a stopping time, $\{N = j\} \in \mathcal{F}_j$. For $j \leq k$, the submartingale property over several steps gives

$$\mathbb{E}(X_k \mid \mathcal{F}_j) \geq X_j.$$

Multiplying by $\mathbf{1}_{\{N=j\}}$ and taking expectations,

$$\mathbb{E}(X_k; N = j) = \mathbb{E}\{\mathbf{1}_{\{N=j\}}\mathbb{E}(X_k \mid \mathcal{F}_j)\} \geq \mathbb{E}(X_j; N = j).$$

Since $N \leq k$ almost surely,

$$\mathbb{E}X_N = \sum_{j=0}^k \mathbb{E}(X_j; N = j) \leq \sum_{j=0}^k \mathbb{E}(X_k; N = j) = \mathbb{E}X_k.$$

Exercise 4.4.2

Question. Generalize Theorem 4.4.1 as follows: if (X_n) is a submartingale and $M \leq N$ are stopping times with $\mathbb{P}(N \leq k) = 1$, prove that

$$\mathbb{E}X_M \leq \mathbb{E}X_N.$$

Knowledge used. `durrett_ch4_bounded_stopping_submartingale_expectation`;
`durrett_ch4_stopped_process_supermartingale`;
`durrett_ch4_martingale_transform_no_gambling_system`.

Solution. Define the predictable process

$$H_m = \mathbf{1}_{\{M < m \leq N\}}, \quad 1 \leq m \leq k.$$

Indeed,

$$\{M < m \leq N\} = \{M \leq m-1\} \cap \{N \geq m\} \in \mathcal{F}_{m-1}.$$

Since $H_m \geq 0$ and X is a submartingale, the martingale-transform principle says that

$$(H \cdot X)_n = \sum_{m=1}^n H_m(X_m - X_{m-1})$$

is a submartingale starting from 0. At time k ,

$$(H \cdot X)_k = X_N - X_M,$$

because the sum includes exactly the increments after M and up to N . Therefore

$$\mathbb{E}X_N - \mathbb{E}X_M = \mathbb{E}(H \cdot X)_k \geq 0,$$

which proves $\mathbb{E}X_M \leq \mathbb{E}X_N$.

Exercise 4.4.3

Question. Suppose $M \leq N$ are stopping times. If $A \in \mathcal{F}_M$, show that

$$L = \begin{cases} M, & \text{on } A, \\ N, & \text{on } A^c \end{cases}$$

is a stopping time.

Knowledge used. `durrett_ch4_stopped_process_supermartingale`;
`durrett_ch4_bounded_stopping_submartingale_expectation`.

Solution. For each n ,

$$\{L \leq n\} = (A \cap \{M \leq n\}) \cup (A^c \cap \{N \leq n\}).$$

Since $A \in \mathcal{F}_M$, the event $A \cap \{M \leq n\}$ belongs to $\mathcal{F}_n$. Also, because $M \leq N$, on $\{N \leq n\}$ we have $M \leq n$, and the definition of $A \in \mathcal{F}_M$ implies

$$A^c \cap \{N \leq n\} = A^c \cap \{M \leq n\} \cap \{N \leq n\} \in \mathcal{F}_n.$$

Therefore $\{L \leq n\} \in \mathcal{F}_n$ for every n , so L is a stopping time.

Exercise 4.4.4

Question. Use Exercise 4.4.3 to strengthen Exercise 4.4.2 to

$$X_M \leq \mathbb{E}(X_N \mid \mathcal{F}_M)$$

for bounded stopping times $M \leq N$.

Knowledge used. `durrett_ch4_bounded_stopping_submartingale_expectation`;
`durrett_ch4_conditional_expectation_definition`;
`durrett_ch4_stopped_process_supermartingale`.

Solution. Let $A \in \mathcal{F}_M$ and define $L = M$ on A and $L = N$ on A^c . By Exercise 4.4.3, L is a stopping time and $M \leq L \leq N$. Exercise 4.4.2 gives

$$\mathbb{E}X_M \leq \mathbb{E}X_L \leq \mathbb{E}X_N.$$

Since

$$X_L = X_M \mathbf{1}_A + X_N \mathbf{1}_{A^c},$$

the inequality $\mathbb{E}X_L \leq \mathbb{E}X_N$ gives

$$\mathbb{E}(X_M; A) + \mathbb{E}(X_N; A^c) \leq \mathbb{E}(X_N; A) + \mathbb{E}(X_N; A^c).$$

Thus

$$\mathbb{E}(X_M; A) \leq \mathbb{E}(X_N; A) \quad \text{for every } A \in \mathcal{F}_M.$$

By the defining property of conditional expectation, this is equivalent to

$$X_M \leq \mathbb{E}(X_N \mid \mathcal{F}_M) \quad \text{a.s.}$$

Exercise 4.4.5

Question. Prove the following conditional-variance identity. If $\mathcal{F} \subseteq \mathcal{G}$, then

$$\mathbb{E}(\mathbb{E}(Y \mid \mathcal{G}) - \mathbb{E}(Y \mid \mathcal{F}))^2 = \mathbb{E}(\mathbb{E}(Y \mid \mathcal{G}))^2 - \mathbb{E}(\mathbb{E}(Y \mid \mathcal{F}))^2.$$

Knowledge used. `durrett_ch4_conditional_expectation_l2_projection`;
`durrett_ch4_tower_property_smaller_sigma_field_wins`; `durrett_ch4_taking_out_what_is_known`.

Solution. Let

$$U = \mathbb{E}(Y \mid \mathcal{G}), \quad V = \mathbb{E}(Y \mid \mathcal{F}).$$

Since $\mathcal{F} \subseteq \mathcal{G}$, the tower property gives

$$\mathbb{E}(U \mid \mathcal{F}) = \mathbb{E}(\mathbb{E}(Y \mid \mathcal{G}) \mid \mathcal{F}) = \mathbb{E}(Y \mid \mathcal{F}) = V.$$

Therefore

$$\mathbb{E}(UV) = \mathbb{E}\{V\mathbb{E}(U \mid \mathcal{F})\} = \mathbb{E}(V^2).$$

Expanding the square,

$$\begin{aligned} \mathbb{E}(U - V)^2 &= \mathbb{E}U^2 - 2\mathbb{E}(UV) + \mathbb{E}V^2 \\ &= \mathbb{E}U^2 - \mathbb{E}V^2, \end{aligned}$$

which is the desired identity.

Exercise 4.4.6

Question. Let $S_n = \xi_1 + \cdots + \xi_n$, where the ξ_m are independent with $\mathbb{E}\xi_m = 0$, $\sigma_m^2 = \mathbb{E}\xi_m^2$, and $|\xi_m| \leq K$. Put

$$s_n^2 = \sum_{m=1}^n \sigma_m^2 = \text{Var}(S_n).$$

Use the martingale $S_n^2 - s_n^2$ and Theorem 4.4.1 to prove

$$\mathbb{P}\left(\max_{1 \leq m \leq n} |S_m| \leq x\right) \leq \frac{(x+K)^2}{\text{Var}(S_n)}.$$

Knowledge used. `durrett_ch4_random_walk_martingales`;
`durrett_ch4_bounded_stopping_submartingale_expectation`;
`durrett_ch4_conditional_variance_formula_martingales`.

Solution. Let

$$T = \inf\{m \geq 1 : |S_m| > x\}.$$

The process $S_m^2 - s_m^2$ is a martingale. Applying bounded optional sampling to $T \wedge n$ gives

$$\mathbb{E}S_{T \wedge n}^2 = \mathbb{E}s_{T \wedge n}^2.$$

On the event

$$A = \left\{ \max_{1 \leq m \leq n} |S_m| \leq x \right\},$$

we have $T > n$, so $s_{T \wedge n}^2 = s_n^2$. Hence

$$\mathbb{E}s_{T \wedge n}^2 \geq s_n^2 \mathbb{P}(A).$$

On the other hand, if $T \leq n$, then the bounded increment assumption gives $|S_T| \leq x + K$, while if $T > n$, $|S_n| \leq x$. Thus

$$|S_{T \wedge n}| \leq x + K \quad \text{always.}$$

Therefore

$$s_n^2 \mathbb{P}(A) \leq \mathbb{E}S_{T \wedge n}^2 \leq (x + K)^2.$$

Since $s_n^2 = \text{Var}(S_n)$, the result follows.

Exercise 4.4.7

Question. Let (X_n) be a martingale with $X_0 = 0$ and $\mathbb{E}X_n^2 < \infty$. Show that

$$\mathbb{P}\left(\max_{1 \leq m \leq n} X_m \geq \lambda\right) \leq \frac{\mathbb{E}X_n^2}{\mathbb{E}X_n^2 + \lambda^2}.$$

Knowledge used. `durrett_ch4_doob_maximal_inequality`;
`durrett_ch4_convex_transform_submartingale`; `durrett_ch4_orthogonal_martingale_increments`.

Solution. For any $c > 0$, the process

$$Y_m = (X_m + c)^2$$

is a nonnegative submartingale, since it is a convex function of a martingale. If

$$A = \left\{ \max_{1 \leq m \leq n} X_m \geq \lambda \right\},$$

then on A the maximum of Y_m is at least $(\lambda + c)^2$. Doob's inequality gives

$$\mathbb{P}(A) \leq \frac{\mathbb{E}Y_n}{(\lambda + c)^2} = \frac{\mathbb{E}X_n^2 + c^2}{(\lambda + c)^2},$$

because $\mathbb{E}X_n = 0$. Let $v = \mathbb{E}X_n^2$. Optimizing over $c > 0$ gives $c = v/\lambda$, and hence

$$\mathbb{P}(A) \leq \frac{v}{v + \lambda^2}.$$

Exercise 4.4.8

Question. Let (X_n) be a submartingale and

$$\bar{X}_n = \max_{0 \leq m \leq n} X_m^+, \quad \log^+ x = \max(\log x, 0).$$

Prove

$$\mathbb{E}\bar{X}_n \leq (1 - e^{-1})^{-1} \left\{ 1 + \mathbb{E} [X_n^+ \log^+(X_n^+)] \right\}.$$

Knowledge used. [durrett_ch4_doob_maximal_inequality](#);
[durrett_ch4_conditional_expectation_properties](#).

Solution. Let $M > 0$. Using the tail integral formula and the trivial bound $\mathbb{P}(\bar{X}_n \wedge M \geq \lambda) \leq 1$ for $\lambda \leq 1$,

$$\begin{aligned} \mathbb{E}(\bar{X}_n \wedge M) &= \int_0^\infty \mathbb{P}(\bar{X}_n \wedge M \geq \lambda) d\lambda \\ &\leq 1 + \int_1^\infty \mathbb{P}(\bar{X}_n \wedge M \geq \lambda) d\lambda. \end{aligned}$$

For $\lambda > 1$, Doob's inequality gives

$$\mathbb{P}(\bar{X}_n \wedge M \geq \lambda) \leq \lambda^{-1} \mathbb{E} [X_n^+; \bar{X}_n \wedge M \geq \lambda].$$

Therefore, by Fubini,

$$\begin{aligned} \mathbb{E}(\bar{X}_n \wedge M) &\leq 1 + \int_1^\infty X_n^+ \int_1^{\bar{X}_n \wedge M} \frac{d\lambda}{\lambda} d\mathbb{P} \\ &\leq 1 + \mathbb{E} [X_n^+ \log(\bar{X}_n \wedge M)]. \end{aligned}$$

The calculus inequality

$$a \log b \leq a \log a + b/e \leq a \log^+ a + b/e, \quad a, b \geq 0, \quad b > 0,$$

follows by maximizing $a \log b - b/e$ as a function of b . Applying it with $a = X_n^+$ and $b = \bar{X}_n \wedge M$ yields

$$\mathbb{E}(\bar{X}_n \wedge M) \leq 1 + \mathbb{E}[X_n^+ \log^+(X_n^+)] + e^{-1} \mathbb{E}(\bar{X}_n \wedge M).$$

Rearranging,

$$\mathbb{E}(\bar{X}_n \wedge M) \leq (1 - e^{-1})^{-1} \left\{ 1 + \mathbb{E}[X_n^+ \log^+(X_n^+)] \right\}.$$

Letting $M \rightarrow \infty$ and using monotone convergence proves the result.

Exercise 4.4.9

Question. Let (X_n) and (Y_n) be square-integrable martingales. Prove

$$\mathbb{E}X_nY_n - \mathbb{E}X_0Y_0 = \sum_{m=1}^n \mathbb{E}[(X_m - X_{m-1})(Y_m - Y_{m-1})].$$

Knowledge used. [durrett_ch4_orthogonal_martingale_increments](#);
[durrett_ch4_conditional_variance_formula_martingales](#).

Solution. Write

$$\Delta X_m = X_m - X_{m-1}, \quad \Delta Y_m = Y_m - Y_{m-1}.$$

Then

$$X_n = X_0 + \sum_{m=1}^n \Delta X_m, \quad Y_n = Y_0 + \sum_{m=1}^n \Delta Y_m.$$

When $i < j$, ΔX_i is $\mathcal{F}_{j-1}$ -measurable and

$$\mathbb{E}(\Delta X_i \Delta Y_j) = \mathbb{E}\{\Delta X_i \mathbb{E}(\Delta Y_j \mid \mathcal{F}_{j-1})\} = 0.$$

Similarly, terms with $j < i$ vanish, and the terms involving X_0 or Y_0 with later martingale increments vanish. Thus only the matching increments remain:

$$\mathbb{E}X_nY_n = \mathbb{E}X_0Y_0 + \sum_{m=1}^n \mathbb{E}(\Delta X_m \Delta Y_m).$$

Exercise 4.4.10

Question. Let (X_n) be a martingale and put

$$\xi_n = X_n - X_{n-1}, \quad n \geq 1.$$

If $\mathbb{E}X_0^2 < \infty$ and

$$\sum_{m=1}^{\infty} \mathbb{E}\xi_m^2 < \infty,$$

show that X_n converges almost surely and in L^2 .

Knowledge used. [durrett_ch4_orthogonal_martingale_increments](#);
[durrett_ch4_lp_martingale_convergence](#);
[durrett_ch4_conditional_variance_formula_martingales](#).

Solution. By orthogonality of martingale increments,

$$\mathbb{E}X_n^2 = \mathbb{E}X_0^2 + \sum_{m=1}^n \mathbb{E}\xi_m^2.$$

The right side is bounded uniformly in n by assumption. Therefore

$$\sup_n \mathbb{E}|X_n|^2 < \infty.$$

The L^p martingale convergence theorem with $p = 2$ implies that X_n converges almost surely and in L^2 to some square-integrable limit X_∞ .

Exercise 4.4.11

Question. Continue with the notation of Exercise 4.4.10. If $b_m \uparrow \infty$ and

$$\sum_{m=1}^{\infty} \frac{\mathbb{E}\xi_m^2}{b_m^2} < \infty,$$

show that

$$\frac{X_n}{b_n} \rightarrow 0 \quad \text{a.s.}$$

In particular, if $\mathbb{E}\xi_n^2 \leq K < \infty$ and $\sum_m b_m^{-2} < \infty$, then $X_n/b_n \rightarrow 0$ a.s.

Knowledge used. [durrett_ch4_orthogonal_martingale_increments](#);
[durrett_ch4_lp_martingale_convergence](#); [durrett_ch4_martingale_strong_law_from_variances](#).

Solution. Define

$$M_n = \sum_{m=1}^n \frac{\xi_m}{b_m}.$$

This is a martingale, and its increments have square expectations $\mathbb{E}\xi_m^2/b_m^2$. By Exercise 4.4.10, (M_n) converges almost surely. Kronecker's lemma now applies to the convergent series

$$\sum_{m=1}^{\infty} \frac{\xi_m}{b_m}.$$

It gives

$$\frac{1}{b_n} \sum_{m=1}^n \xi_m \rightarrow 0 \quad \text{a.s.}$$

Since

$$\frac{X_n}{b_n} = \frac{X_0}{b_n} + \frac{1}{b_n} \sum_{m=1}^n \xi_m$$

and $b_n \rightarrow \infty$, we conclude that $X_n/b_n \rightarrow 0$ almost surely. The final assertion follows immediately from

$$\sum_m \frac{\mathbb{E}\xi_m^2}{b_m^2} \leq K \sum_m b_m^{-2} < \infty.$$

Exercise 4.6.1

Question. Let $Z_1, Z_2, \dots$ be i.i.d. with $\mathbb{E}|Z_i| < \infty$, let θ be an independent random variable with finite mean, and set

$$Y_i = Z_i + \theta.$$

Show that

$$\mathbb{E}(\theta \mid Y_1, \dots, Y_n) \rightarrow \theta \quad \text{a.s.}$$

Knowledge used. [durrett_ch4_levy_upward_theorem](#);
[durrett_ch4_closed_martingales_ui_equivalence](#); [durrett_ch4_slln_via_backward_martingales](#).

Solution. Let

$$\mathcal{F}_n = \sigma(Y_1, \dots, Y_n), \quad \mathcal{F}_\infty = \sigma(Y_1, Y_2, \dots).$$

By the strong law of large numbers,

$$\frac{1}{n} \sum_{i=1}^n Y_i = \theta + \frac{1}{n} \sum_{i=1}^n Z_i \rightarrow \theta + \mathbb{E}Z_1 \quad \text{a.s.}$$

The left side is $\mathcal{F}_\infty$ -measurable, so θ is $\mathcal{F}_\infty$ -measurable. Hence

$$\mathbb{E}(\theta \mid \mathcal{F}_\infty) = \theta.$$

Since $\mathcal{F}_n \uparrow \mathcal{F}_\infty$, Levy's upward theorem gives

$$\mathbb{E}(\theta \mid Y_1, \dots, Y_n) = \mathbb{E}(\theta \mid \mathcal{F}_n) \rightarrow \mathbb{E}(\theta \mid \mathcal{F}_\infty) = \theta \quad \text{a.s.}$$

Exercise 4.6.2

Question. Let $\Omega = [0, 1)$,

$$I_{k,n} = [k2^{-n}, (k+1)2^{-n}), \quad 0 \leq k < 2^n,$$

and $\mathcal{F}_n = \sigma(I_{k,n} : 0 \leq k < 2^n)$. Suppose f is Lipschitz on $[0, 1)$, so $|f(t) - f(s)| \leq K|t - s|$. Define

$$X_n(\omega) = \frac{f((k+1)2^{-n}) - f(k2^{-n})}{2^{-n}} \quad \text{for } \omega \in I_{k,n}.$$

Show that (X_n) is a martingale, that $X_n \rightarrow X_\infty$ a.s. and in L^1 , and that

$$f(b) - f(a) = \int_a^b X_\infty(\omega) d\omega.$$

Knowledge used. [durrett_ch4_martingale_definition](#);
[durrett_ch4_closed_martingales_ui_equivalence](#); [durrett_ch4_levy_upward_theorem](#).

Solution. The random variable X_n is $\mathcal{F}_n$ -measurable and $|X_n| \leq K$. On a dyadic parent interval

$$I = [k2^{-n}, (k+1)2^{-n}),$$

the two children at level $n+1$ have slopes whose average is

$$\begin{aligned} & \frac{1}{2} \frac{f((2k+1)2^{-(n+1)}) - f(2k2^{-(n+1)})}{2^{-(n+1)}} \\ & + \frac{1}{2} \frac{f((2k+2)2^{-(n+1)}) - f((2k+1)2^{-(n+1)})}{2^{-(n+1)}} \\ & = \frac{f((k+1)2^{-n}) - f(k2^{-n})}{2^{-n}}. \end{aligned}$$

Thus $\mathbb{E}(X_{n+1} \mid \mathcal{F}_n) = X_n$, so (X_n) is a martingale.

Since $|X_n| \leq K$, the martingale is uniformly integrable. The martingale convergence theorem for uniformly integrable martingales implies

$$X_n \rightarrow X_\infty \quad \text{a.s. and in } L^1.$$

For dyadic $a < b$, the definition of X_n gives, for all sufficiently large n ,

$$\int_a^b X_n(\omega) d\omega = f(b) - f(a).$$

Letting $n \rightarrow \infty$ and using L^1 convergence gives the formula for dyadic a, b . For general $a < b$, choose dyadic $a_m \downarrow a$ and $b_m \uparrow b$ with $a_m, b_m \in [0, 1)$. The Lipschitz continuity of f and integrability of X_∞ allow passage to the limit, giving

$$f(b) - f(a) = \int_a^b X_\infty(\omega) d\omega.$$

Exercise 4.6.3

Question. With the dyadic sigma-fields from Exercise 4.6.2, suppose f is integrable on $[0, 1)$. Show that $\mathbb{E}(f \mid \mathcal{F}_n)$ is a step function and

$$\mathbb{E}(f \mid \mathcal{F}_n) \rightarrow f \quad \text{in } L^1.$$

Conclude that every integrable f can be approximated in L^1 by a step function.

Knowledge used. [durrett_ch4_levy_upward_theorem](#);
[durrett_ch4_partition_conditioning_formula](#); [durrett_ch4_ui_plus_probability_equals_11](#).

Solution. On each dyadic atom $I_{k,n}$,

$$\mathbb{E}(f \mid \mathcal{F}_n) = 2^n \int_{I_{k,n}} f(x) dx.$$

Thus $\mathbb{E}(f \mid \mathcal{F}_n)$ is a dyadic step function. The dyadic sigma-fields increase to the Borel sigma-field on $[0, 1)$, completed if necessary. By Levy's upward theorem,

$$\mathbb{E}(f \mid \mathcal{F}_n) \rightarrow \mathbb{E}(f \mid \mathcal{F}_\infty) = f \quad \text{a.s. and in } L^1.$$

Therefore, given $\varepsilon > 0$, choose n large enough so that

$$\int_0^1 |\mathbb{E}(f \mid \mathcal{F}_n) - f| dx < \varepsilon.$$

The function $\mathbb{E}(f \mid \mathcal{F}_n)$ is the desired step-function approximation.

Exercise 4.6.4

Question. Let X_n take values in $[0, \infty)$, and set

$$D = \{X_n = 0 \text{ for some } n \geq 1\}.$$

Assume that for every $x < \infty$ there is a number $\delta(x) > 0$ such that

$$\mathbb{P}(D \mid X_1, \dots, X_n) \geq \delta(x) \quad \text{a.s. on } \{X_n \leq x\}.$$

Use Levy's 0-1 law to prove

$$\mathbb{P}\left(D \cup \left\{\lim_{n \rightarrow \infty} X_n = \infty\right\}\right) = 1.$$

Knowledge used. [durrett_ch4_levy_zero_one_law](#); [durrett_ch4_levy_upward_theorem](#); [durrett_ch4_conditional_expectation_definition](#).

Solution. Let $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$ and $\mathcal{F}_\infty = \sigma(X_1, X_2, \dots)$. Since $D \in \mathcal{F}_\infty$, Levy's 0-1 law gives

$$\mathbb{P}(D \mid \mathcal{F}_n) = \mathbb{E}(\mathbf{1}_D \mid \mathcal{F}_n) \rightarrow \mathbf{1}_D \quad \text{a.s.}$$

Now suppose that D does not occur and that X_n does not tend to ∞ . Then for some finite x there are infinitely many n with $X_n \leq x$. Along this subsequence,

$$\mathbb{P}(D \mid \mathcal{F}_n) \geq \delta(x) > 0.$$

Taking the subsequential limit gives $\mathbf{1}_D \geq \delta(x)$, impossible on D^c . Hence, outside a null set, D^c implies $X_n \rightarrow \infty$. This is equivalent to

$$\mathbb{P}\left(D \cup \{\lim_n X_n = \infty\}\right) = 1.$$

Exercise 4.6.5

Question. Let (Z_n) be a branching process with offspring distribution (p_k) , and assume $p_0 > 0$. Use Exercise 4.6.4 to show that

$$\mathbb{P}\left(\lim_{n \rightarrow \infty} Z_n = 0 \text{ or } \infty\right) = 1.$$

Knowledge used. [durrett_ch4_branching_process_normalized_martingale](#); [durrett_ch4_levy_zero_one_law](#).

Solution. Let

$$D = \{Z_n = 0 \text{ for some } n\}.$$

Once a branching process hits 0, it remains at 0. If $Z_n = k \leq x$, then all k individuals in generation n can have no children in the next generation, an event of conditional probability at least p_0^k . Thus, on $\{Z_n \leq x\}$,

$$\mathbb{P}(D \mid Z_0, \dots, Z_n) \geq p_0^{\lfloor x \rfloor} > 0.$$

Exercise 4.6.4 gives

$$\mathbb{P}\left(D \cup \{\lim_n Z_n = \infty\}\right) = 1.$$

On D , $Z_n = 0$ for all sufficiently large n , so $\lim_n Z_n = 0$. Therefore

$$\mathbb{P}\left(\lim_n Z_n = 0 \text{ or } \infty\right) = 1.$$

Exercise 4.6.6

Question. Let $X_n \in [0, 1]$ be adapted to $\mathcal{F}_n$. Let $\alpha, \beta > 0$ with $\alpha + \beta = 1$, and suppose

$$\mathbb{P}(X_{n+1} = \alpha + \beta X_n \mid \mathcal{F}_n) = X_n, \quad \mathbb{P}(X_{n+1} = \beta X_n \mid \mathcal{F}_n) = 1 - X_n.$$

Show that

$$\mathbb{P}(\lim_n X_n = 0 \text{ or } 1) = 1,$$

and if $X_0 = \theta$, then

$$\mathbb{P}(\lim_n X_n = 1) = \theta.$$

Knowledge used. [durrett_ch4_martingale_definition](#);
[durrett_ch4_closed_martingales_ui_equivalence](#);
[durrett_ch4_conditional_variance_formula_martingales](#).

Solution. First compute the conditional mean:

$$\begin{aligned}\mathbb{E}(X_{n+1} \mid \mathcal{F}_n) &= X_n(\alpha + \beta X_n) + (1 - X_n)\beta X_n \\ &= \alpha X_n + \beta X_n = X_n.\end{aligned}$$

Thus (X_n) is a bounded martingale, so $X_n \rightarrow X_\infty$ a.s. and in L^2 .

The increments are

$$X_{n+1} - X_n = \begin{cases} \alpha(1 - X_n), & \text{with conditional probability } X_n, \\ -\alpha X_n, & \text{with conditional probability } 1 - X_n. \end{cases}$$

Hence

$$\mathbb{E}((X_{n+1} - X_n)^2 \mid \mathcal{F}_n) = \alpha^2 X_n(1 - X_n).$$

Since X_n converges in L^2 , the left side has expectations tending to 0, and bounded convergence gives

$$0 = \alpha^2 \mathbb{E}[X_\infty(1 - X_\infty)].$$

Thus $X_\infty(1 - X_\infty) = 0$ a.s., so $X_\infty \in \{0, 1\}$ a.s.

Finally, $\mathbb{E}X_n = \mathbb{E}X_0 = \theta$ for all n , and bounded convergence gives

$$\theta = \mathbb{E}X_\infty = \mathbb{P}(X_\infty = 1).$$

Exercise 4.6.7

Question. Show that if $\mathcal{F}_n \uparrow \mathcal{F}_\infty$ and $Y_n \rightarrow Y$ in L^1 , then

$$\mathbb{E}(Y_n \mid \mathcal{F}_n) \rightarrow \mathbb{E}(Y \mid \mathcal{F}_\infty) \quad \text{in } L^1.$$

Knowledge used. [durrett_ch4_levy_upward_theorem](#);
[durrett_ch4_conditional_jensen_contraction](#); [durrett_ch4_ui_plus_probability_equals_11](#).

Solution. By the L^1 contraction property of conditional expectation,

$$\begin{aligned}\mathbb{E}|\mathbb{E}(Y_n \mid \mathcal{F}_n) - \mathbb{E}(Y \mid \mathcal{F}_n)| &\leq \mathbb{E}(\mathbb{E}(|Y_n - Y| \mid \mathcal{F}_n)) \\ &= \mathbb{E}|Y_n - Y| \rightarrow 0.\end{aligned}$$

By Levy's upward theorem,

$$\mathbb{E}(Y \mid \mathcal{F}_n) \rightarrow \mathbb{E}(Y \mid \mathcal{F}_\infty) \quad \text{in } L^1.$$

The triangle inequality combines the two limits and proves

$$\mathbb{E}(Y_n \mid \mathcal{F}_n) \rightarrow \mathbb{E}(Y \mid \mathcal{F}_\infty) \quad \text{in } L^1.$$

Exercise 4.7.1

Question. Show that if a backwards martingale has $X_0 \in L^p$, then the convergence in Theorem 4.7.1 occurs in L^p .

Knowledge used. `durrett_ch4_backward_martingale_convergence`;
`durrett_ch4_levy_downward_theorem`; `durrett_ch4_conditional_jensen_contraction`;
`durrett_ch4_uniform_integrability_definition`.

Solution. Let $(X_n)_{n \leq 0}$ be a backwards martingale. Then

$$X_n = \mathbb{E}(X_0 \mid \mathcal{F}_n), \quad n \leq 0.$$

By Theorem 4.7.1 and Theorem 4.7.2,

$$X_n \rightarrow X_{-\infty} := \mathbb{E}(X_0 \mid \mathcal{F}_{-\infty}) \quad \text{a.s. and in } L^1.$$

Assume $p \geq 1$ and $X_0 \in L^p$. Conditional Jensen gives

$$|X_n|^p \leq \mathbb{E}(|X_0|^p \mid \mathcal{F}_n).$$

The family on the right is uniformly integrable by the conditional expectation uniform-integrability theorem, applied to $|X_0|^p \in L^1$. Also $X_{-\infty} \in L^p$ by the L^p contraction property. Hence

$$|X_n - X_{-\infty}|^p \leq 2^{p-1}(|X_n|^p + |X_{-\infty}|^p)$$

is a uniformly integrable family. Since $X_n \rightarrow X_{-\infty}$ in probability, uniform integrability implies

$$\mathbb{E}|X_n - X_{-\infty}|^p \rightarrow 0.$$

Thus the convergence occurs in L^p .

Exercise 4.7.2

Question. Prove the backwards analogue of Theorem 4.6.10. Suppose $Y_n \rightarrow Y_{-\infty}$ a.s. as $n \rightarrow -\infty$ and $|Y_n| \leq Z$ for all n , where $\mathbb{E}Z < \infty$. If $\mathcal{F}_n \downarrow \mathcal{F}_{-\infty}$ as $n \rightarrow -\infty$, prove

$$\mathbb{E}(Y_n \mid \mathcal{F}_n) \rightarrow \mathbb{E}(Y_{-\infty} \mid \mathcal{F}_{-\infty}) \quad \text{a.s.}$$

Knowledge used. `durrett_ch4_levy_downward_theorem`;
`durrett_ch4_conditional_expectation_properties`;
`durrett_ch4_conditional_jensen_contraction`.

Solution. For $N \geq 1$, define

$$W_N = \sup_{m \leq -N} |Y_m - Y_{-\infty}|.$$

Then $0 \leq W_N \leq 2Z$ and $W_N \downarrow 0$ a.s. as $N \rightarrow \infty$. For $n \leq -N$,

$$\begin{aligned} |\mathbb{E}(Y_n \mid \mathcal{F}_n) - \mathbb{E}(Y_{-\infty} \mid \mathcal{F}_n)| &\leq \mathbb{E}(|Y_n - Y_{-\infty}| \mid \mathcal{F}_n) \\ &\leq \mathbb{E}(W_N \mid \mathcal{F}_n). \end{aligned}$$

By Levy's downward theorem,

$$\mathbb{E}(W_N \mid \mathcal{F}_n) \rightarrow \mathbb{E}(W_N \mid \mathcal{F}_{-\infty}) \quad \text{a.s. as } n \rightarrow -\infty.$$

Since $W_N \downarrow 0$ and $W_N \leq 2Z$, conditional monotone convergence gives

$$\mathbb{E}(W_N \mid \mathcal{F}_{-\infty}) \downarrow 0.$$

It follows that

$$\mathbb{E}(Y_n \mid \mathcal{F}_n) - \mathbb{E}(Y_{-\infty} \mid \mathcal{F}_n) \rightarrow 0 \quad \text{a.s.}$$

A second application of Levy's downward theorem gives

$$\mathbb{E}(Y_{-\infty} \mid \mathcal{F}_n) \rightarrow \mathbb{E}(Y_{-\infty} \mid \mathcal{F}_{-\infty}) \quad \text{a.s.}$$

The triangle inequality proves the claim.

Exercise 4.7.3

Question. Prove directly from the definition that if $X_1, X_2, \dots \in \{0, 1\}$ are exchangeable and $S_n = X_1 + \dots + X_n$, then

$$\mathbb{P}(X_1 = 1, \dots, X_k = 1 \mid S_n = m) = \frac{\binom{n-k}{n-m}}{\binom{n}{m}}.$$

Knowledge used. [durrett_ch4_definetti_theorem](#);
[durrett_ch4_backward_martingale_convergence](#).

Solution. Condition on the event $\{S_n = m\}$. By exchangeability, every binary string of length n with exactly m ones has the same conditional probability. There are $\binom{n}{m}$ such strings.

The event $\{X_1 = \dots = X_k = 1\}$ can occur only if the remaining $n - k$ coordinates contain $m - k$ additional ones. The number of such strings is

$$\binom{n-k}{m-k} = \binom{n-k}{n-m}.$$

Therefore

$$\mathbb{P}(X_1 = 1, \dots, X_k = 1 \mid S_n = m) = \frac{\binom{n-k}{m-k}}{\binom{n}{m}} = \frac{\binom{n-k}{n-m}}{\binom{n}{m}}.$$

Exercise 4.7.4

Question. If $X_1, X_2, \dots \in \mathbb{R}$ are exchangeable and $\mathbb{E}X_i^2 < \infty$, prove

$$\mathbb{E}(X_1 X_2) \geq 0.$$

Knowledge used. [durrett_ch4_definetti_theorem](#);
[durrett_ch4_orthogonal_martingale_increments](#).

Solution. For each n ,

$$0 \leq \mathbb{E} \left(\sum_{i=1}^n X_i \right)^2.$$

By exchangeability,

$$\mathbb{E} \left(\sum_{i=1}^n X_i \right)^2 = n\mathbb{E}X_1^2 + n(n-1)\mathbb{E}(X_1 X_2).$$

Thus

$$\mathbb{E}(X_1 X_2) \geq -\frac{\mathbb{E}X_1^2}{n-1} \quad \text{for every } n \geq 2.$$

Letting $n \rightarrow \infty$ gives $\mathbb{E}(X_1 X_2) \geq 0$.

Exercise 4.7.5

Question. Use the first few lines of the proof of Lemma 4.7.7 to conclude that if $X_1, X_2, \dots$ are i.i.d. with $\mathbb{E}X_i = \mu$ and $\text{Var}(X_i) = \sigma^2 < \infty$, then

$$\binom{n}{2}^{-1} \sum_{1 \leq i < j \leq n} (X_i - X_j)^2 \rightarrow 2\sigma^2 \quad \text{a.s.}$$

Knowledge used. [durrett_ch4_backward_martingale_convergence](#);
[durrett_ch4_slln_via_backward_martingales](#); [durrett_ch4_hewitt_savage_zero_one](#).

Solution. Let

$$U_n = \binom{n}{2}^{-1} \sum_{1 \leq i < j \leq n} (X_i - X_j)^2.$$

This is the unordered version of the symmetric average in Lemma 4.7.7 with $\varphi(x, y) = (x - y)^2$. Since $\mathbb{E}\varphi(X_1, X_2) < \infty$, the bounded case in Lemma 4.7.7 applies first to $\varphi_M = \varphi \wedge M$, and then monotone/truncation estimates let $M \rightarrow \infty$. Hence

$$U_n \rightarrow \mathbb{E}[(X_1 - X_2)^2] \quad \text{a.s.}$$

For independent copies with common mean μ and variance σ^2 ,

$$\mathbb{E}[(X_1 - X_2)^2] = \mathbb{E}X_1^2 + \mathbb{E}X_2^2 - 2\mathbb{E}X_1\mathbb{E}X_2 = 2(\sigma^2 + \mu^2) - 2\mu^2 = 2\sigma^2.$$

Therefore $U_n \rightarrow 2\sigma^2$ almost surely.

Exercise 4.8.1

Question. Generalize Theorem 4.8.2. Suppose $L \leq M$ are stopping times and $(Y_{M \wedge n})$ is a uniformly integrable submartingale. Show that

$$\mathbb{E}Y_L \leq \mathbb{E}Y_M \quad \text{and} \quad Y_L \leq \mathbb{E}(Y_M \mid \mathcal{F}_L).$$

Knowledge used. [durrett_ch4_optional_stopping_ui](#);
[durrett_ch4_optional_stopping_integrable_terminal](#);
[durrett_ch4_bounded_stopping_submartingale_expectation](#).

Solution. For bounded truncations, $L \wedge n \leq M \wedge n$, so bounded optional sampling gives

$$\mathbb{E}Y_{L \wedge n} \leq \mathbb{E}Y_{M \wedge n}.$$

Uniform integrability of $(Y_{M \wedge n})$ implies $Y_{M \wedge n} \rightarrow Y_M$ in L^1 ; the same stopped-submartingale argument gives $Y_{L \wedge n} \rightarrow Y_L$ in L^1 . Letting $n \rightarrow \infty$ yields

$$\mathbb{E}Y_L \leq \mathbb{E}Y_M.$$

For the conditional form, let $A \in \mathcal{F}_L$ and define $R = L$ on A and $R = M$ on A^c . Then R is a stopping time with $L \leq R \leq M$, so

$$\mathbb{E}Y_R \leq \mathbb{E}Y_M.$$

Since $Y_R = Y_L \mathbf{1}_A + Y_M \mathbf{1}_{A^c}$,

$$\mathbb{E}(Y_L; A) \leq \mathbb{E}(Y_M; A), \quad A \in \mathcal{F}_L.$$

This is exactly

$$Y_L \leq \mathbb{E}(Y_M \mid \mathcal{F}_L) \quad \text{a.s.}$$

Exercise 4.8.2

Question. If (X_n) is a nonnegative supermartingale, prove

$$\mathbb{P}\left(\sup_{n \geq 0} X_n > \lambda\right) \leq \frac{\mathbb{E}X_0}{\lambda}.$$

Knowledge used. `durrett_ch4_nonnegative_supermartingale_convergence;`
`durrett_ch4_doob_maximal_inequality;` `durrett_ch4_stopped_process_supermartingale.`

Solution. Let $T = \inf\{n : X_n > \lambda\}$. For each n , the stopped process $X_{T \wedge n}$ is a nonnegative supermartingale, so

$$\mathbb{E}X_0 \geq \mathbb{E}X_{T \wedge n} \geq \mathbb{E}(X_T; T \leq n) \geq \lambda \mathbb{P}(T \leq n).$$

Letting $n \rightarrow \infty$ gives

$$\mathbb{P}\left(\sup_{n \geq 0} X_n > \lambda\right) = \mathbb{P}(T < \infty) \leq \frac{\mathbb{E}X_0}{\lambda}.$$

Exercise 4.8.3

Question. Let $S_n = \xi_1 + \cdots + \xi_n$, where the ξ_i are independent with $\mathbb{E}\xi_i = 0$ and $\text{Var}(\xi_i) = \sigma^2$. Let

$$T = \min\{n : |S_n| > a\}.$$

Use the martingale $S_n^2 - n\sigma^2$ to show

$$\mathbb{E}T \geq a^2/\sigma^2.$$

Knowledge used. `durrett_ch4_random_walk_martingales;`
`durrett_ch4_optional_stopping_integrable_terminal;`
`durrett_ch4_conditional_variance_formula_martingales.`

Solution. For the bounded stopping time $T \wedge n$,

$$\mathbb{E}S_{T \wedge n}^2 = \sigma^2 \mathbb{E}(T \wedge n).$$

For a nondegenerate centered random walk, $T < \infty$ almost surely. Fatou's lemma gives

$$\mathbb{E}S_T^2 \leq \liminf_{n \rightarrow \infty} \mathbb{E}S_{T \wedge n}^2 = \sigma^2 \mathbb{E}T.$$

Since $|S_T| > a$, we have $\mathbb{E}S_T^2 \geq a^2$, so

$$\mathbb{E}T \geq a^2/\sigma^2.$$

Exercise 4.8.4

Question. Wald's second equation. Let $S_n = \xi_1 + \cdots + \xi_n$, where the ξ_i are independent with $\mathbb{E}\xi_i = 0$ and $\text{Var}(\xi_i) = \sigma^2$. If T is a stopping time with $\mathbb{E}T < \infty$, prove

$$\mathbb{E}S_T^2 = \sigma^2 \mathbb{E}T.$$

Knowledge used. `durrett_ch4_wald_equation;` `durrett_ch4_random_walk_martingales;`
`durrett_ch4_orthogonal_martingale_increments.`

Solution. For $T \wedge n$,

$$\mathbb{E}S_{T \wedge n}^2 = \sigma^2 \mathbb{E}(T \wedge n).$$

Since $\mathbb{E}T < \infty$, $\mathbb{E}(T \wedge n) \rightarrow \mathbb{E}T$. Also

$$S_T - S_{T \wedge n} = \sum_{m=n+1}^T \xi_m$$

on $\{T > n\}$, and martingale-increment orthogonality gives

$$\mathbb{E}(S_T - S_{T \wedge n})^2 = \sigma^2 \mathbb{E}(T - T \wedge n) \rightarrow 0.$$

Thus $S_{T \wedge n} \rightarrow S_T$ in L^2 , and

$$\mathbb{E}S_T^2 = \lim_{n \rightarrow \infty} \mathbb{E}S_{T \wedge n}^2 = \sigma^2 \mathbb{E}T.$$

Exercise 4.8.5

Question. Let $P(\xi_i = 1) = p$ and $P(\xi_i = -1) = q = 1 - p$, where $p < 1/2$. Let $S_n = S_0 + \xi_1 + \dots + \xi_n$ and

$$V_0 = \min\{n \geq 0 : S_n = 0\}.$$

Given $\mathbb{E}_x V_0 = x/(1 - 2p)$, compute $\text{Var}_x(V_0)$.

Knowledge used. [durrett_ch4_asymmetric_gamblers_ruin](#); [durrett_ch4_wald_equation](#); [durrett_ch4_random_walk_martingales](#).

Solution. Let $d = q - p = 1 - 2p > 0$ and $\tau = V_0$. The centered increment $Y_i = \xi_i - (p - q) = \xi_i + d$ has variance

$$\text{Var}(Y_i) = \text{Var}(\xi_i) = 1 - (p - q)^2 = 1 - d^2.$$

Starting from $S_0 = x$,

$$(S_n - x - (p - q)n)^2 - n(1 - d^2)$$

is a martingale. At τ , $S_\tau = 0$, so optional stopping gives

$$\mathbb{E}(-x + d\tau)^2 = (1 - d^2)\mathbb{E}\tau.$$

Using $\mathbb{E}\tau = x/d$,

$$d^2 \mathbb{E}\tau^2 - 2xd\mathbb{E}\tau + x^2 = (1 - d^2)\frac{x}{d},$$

hence

$$\mathbb{E}\tau^2 = \frac{x^2}{d^2} + \frac{x(1 - d^2)}{d^3}.$$

Therefore

$$\text{Var}_x(V_0) = \frac{x(1 - d^2)}{d^3} = x \frac{1 - (p - q)^2}{(q - p)^3}.$$

The answer must be of the form cx because the time to go from x to 0 is the sum of x independent copies of the time to go down one level.

Exercise 4.8.6

Question. Continue with the setup of Exercise 4.8.5. Find the generating function of V_0 .

Knowledge used. [durrett_ch4_asymmetric_gamblers_ruin](#);
[durrett_ch4_random_walk_martingales](#); [durrett_ch4_optional_stopping_integrable_terminal](#).

Solution. Let

$$\phi(\theta) = \mathbb{E}e^{\theta\xi_1} = pe^\theta + qe^{-\theta}.$$

The exponential martingale gives, for $\theta \leq 0$,

$$e^{\theta x} = \mathbb{E}_x [\phi(\theta)^{-V_0}].$$

For $0 < s < 1$, choose $\theta \leq 0$ with $\phi(\theta) = 1/s$. Writing $y = e^\theta$, solve

$$py + \frac{q}{y} = \frac{1}{s}.$$

The root in $(0, 1]$ is

$$y = \frac{1 - \sqrt{1 - 4pqs^2}}{2ps}.$$

Thus

$$\mathbb{E}_x(s^{V_0}) = \left(\frac{1 - \sqrt{1 - 4pqs^2}}{2ps} \right)^x.$$

It has the form $f(s)^x$ because the descent from x to 0 is a sum of x independent one-level descent times.

Exercise 4.8.7

Question. Let (S_n) be a symmetric simple random walk starting at 0, and let

$$T = \inf\{n : S_n \notin (-a, a)\},$$

where a is a positive integer. Find constants b, c such that

$$Y_n = S_n^4 - 6nS_n^2 + bn^2 + cn$$

is a martingale, and use this to compute $\mathbb{E}T^2$.

Knowledge used. [durrett_ch4_symmetric_gamblers_ruin](#);
[durrett_ch4_random_walk_martingales](#); [durrett_ch4_optional_stopping_integrable_terminal](#).

Solution. For a symmetric ± 1 increment,

$$\mathbb{E}(S_n + \xi_{n+1})^2 \mid \mathcal{F}_n = S_n^2 + 1, \quad \mathbb{E}(S_n + \xi_{n+1})^4 \mid \mathcal{F}_n = S_n^4 + 6S_n^2 + 1.$$

Substitution into $\mathbb{E}(Y_{n+1} \mid \mathcal{F}_n) = Y_n$ gives

$$-6 + 2b = 0, \quad 1 - 6 + b + c = 0.$$

Hence

$$b = 3, \quad c = 2.$$

At exit, $S_T = \pm a$. Also $\mathbb{E}T = a^2$. Optional stopping gives $\mathbb{E}Y_T = Y_0 = 0$, so

$$0 = a^4 - 6a^2\mathbb{E}T + 3\mathbb{E}T^2 + 2\mathbb{E}T.$$

Therefore

$$\mathbb{E}T^2 = \frac{5a^4 - 2a^2}{3}.$$

Exercise 4.8.8

Question. Let $S_n = \xi_1 + \cdots + \xi_n$ be a random walk. Suppose

$$\phi(\theta_0) = \mathbb{E}e^{\theta_0\xi_1} = 1$$

for some $\theta_0 < 0$, and that ξ_i is not constant. Let

$$\tau = \inf\{n : S_n \notin (a, b)\}.$$

Use $X_n = e^{\theta_0 S_n}$ to show

$$\mathbb{E}X_\tau = 1, \quad \mathbb{P}(S_\tau \leq a) \leq e^{-\theta_0 a}.$$

Knowledge used. [durrett_ch4_insurance_ruin_exponential_bound](#);
[durrett_ch4_random_walk_martingales](#); [durrett_ch4_optional_stopping_integrable_terminal](#).

Solution. Since $\phi(\theta_0) = 1$, $X_n = e^{\theta_0 S_n}$ is a nonnegative martingale with $X_0 = 1$. Optional stopping for the two-sided exit time gives

$$\mathbb{E}X_\tau = \mathbb{E}X_0 = 1.$$

On $\{S_\tau \leq a\}$, $\theta_0 < 0$ implies

$$X_\tau = e^{\theta_0 S_\tau} \geq e^{\theta_0 a}.$$

Thus

$$1 = \mathbb{E}X_\tau \geq e^{\theta_0 a} \mathbb{P}(S_\tau \leq a),$$

or

$$\mathbb{P}(S_\tau \leq a) \leq e^{-\theta_0 a}.$$

Exercise 4.8.9

Question. Continue with Exercise 4.8.8. Suppose the ξ_i are integer-valued,

$$\mathbb{P}(\xi_i < -1) = 0, \quad \mathbb{P}(\xi_i = -1) > 0, \quad \mathbb{E}\xi_i > 0.$$

Let $T = \inf\{n : S_n = a\}$, where $a < 0$. Show

$$\mathbb{P}(T < \infty) = e^{-\theta_0 a}.$$

Knowledge used. [durrett_ch4_insurance_ruin_exponential_bound](#);
[durrett_ch4_asymmetric_gamblers_ruin](#); [durrett_ch4_random_walk_martingales](#).

Solution. Because downward jumps are never less than -1 , a lower crossing of a hits a exactly. Let $\tau_b = T_a \wedge T_b$, where $b > 0$. Optional stopping gives

$$1 = \mathbb{E}e^{\theta_0 S_{\tau_b}} = e^{\theta_0 a} \mathbb{P}(T_a < T_b) + \mathbb{E}(e^{\theta_0 S_{T_b}}; T_b < T_a).$$

On $\{T_b < T_a\}$, $S_{T_b} \geq b$, so $e^{\theta_0 S_{T_b}} \leq e^{\theta_0 b} \rightarrow 0$. Letting $b \rightarrow \infty$,

$$1 = e^{\theta_0 a} \mathbb{P}(T_a < \infty),$$

which proves the formula.

Exercise 4.8.10

Question. Consider a favorable game in which the payoffs are -1 , 1 , or 2 , each with probability $1/3$. Starting with fortune i , compute the probability of ever going broke.

Knowledge used. [durrett_ch4_insurance_ruin_exponential_bound](#);
[durrett_ch4_asymmetric_gamblers_ruin](#).

Solution. Let $r = e^\theta \in (0, 1)$. The adjustment equation is

$$\mathbb{E}e^{\theta\xi} = 1 \iff \frac{r^{-1} + r + r^2}{3} = 1.$$

Equivalently,

$$r^3 + r^2 - 3r + 1 = (r - 1)(r^2 + 2r - 1) = 0.$$

The root in $(0, 1)$ is $r = \sqrt{2} - 1$. Starting with fortune i , going broke means the cumulative payoff random walk ever hits $-i$. Hence

$$\mathbb{P}_i(\text{ever go broke}) = (\sqrt{2} - 1)^i.$$

Exercise 4.8.11

Question. Let S_n be an insurance company's assets. In year n , premiums totaling $c > 0$ are received and claims ζ_n are paid, where ζ_n is normal with mean μ and variance σ^2 , with $\mu < c$. If $S_0 > 0$ is nonrandom, show

$$\mathbb{P}(\text{ruin}) \leq \exp\{-2(c - \mu)S_0/\sigma^2\}.$$

Knowledge used. [durrett_ch4_insurance_ruin_exponential_bound](#);
[durrett_ch4_random_walk_martingales](#).

Solution. The increment $\xi_n = c - \zeta_n$ is normal with mean $m = c - \mu > 0$ and variance σ^2 . Its moment generating function is

$$\mathbb{E}e^{\theta\xi_n} = \exp\left(\theta m + \frac{\sigma^2\theta^2}{2}\right).$$

The nonzero solution of $\mathbb{E}e^{\theta\xi_n} = 1$ is

$$\theta_0 = -\frac{2m}{\sigma^2} = -\frac{2(c - \mu)}{\sigma^2}.$$

Ruin means the accumulated-gain random walk, started from 0, falls to $-S_0$ or below. Exercise 4.8.8 with $a = -S_0$ gives

$$\mathbb{P}(\text{ruin}) \leq \exp(\theta_0 S_0) = \exp\{-2(c - \mu)S_0/\sigma^2\}.$$

Exercise 4.9.1

Question. Let $a \in S$,

$$T_a = \inf\{n \geq 1 : X_n = a\},$$

and define

$$f_n = P_a(T_a = n), \quad u_n = P_a(X_n = a).$$

(i) Show that, for $n \geq 1$,

$$u_n = \sum_{m=1}^n f_m u_{n-m}.$$

(ii) Let

$$u(s) = \sum_{n \geq 0} u_n s^n, \quad f(s) = \sum_{n \geq 1} f_n s^n.$$

Show that

$$u(s) = \frac{1}{1 - f(s)}.$$

Knowledge used. [durrett_ch4_arcsine_laws_random_walk](#); [durrett_ch4_reflection_principle](#); [durrett_ch4_ballot_theorem](#).

Solution. Since the chain starts at a , $u_0 = P_a(X_0 = a) = 1$. For $n \geq 1$, decompose the event $\{X_n = a\}$ according to the first positive return time to a :

$$\{X_n = a\} = \bigcup_{m=1}^n \{T_a = m, X_n = a\},$$

where the union is disjoint. By the Markov property at time T_a , on the event $\{T_a = m\}$ the process restarts from a , so

$$P_a(X_n = a \mid T_a = m) = P_a(X_{n-m} = a) = u_{n-m}.$$

Therefore

$$u_n = \sum_{m=1}^n P_a(T_a = m) P_a(X_{n-m} = a) = \sum_{m=1}^n f_m u_{n-m}.$$

Now multiply by s^n and sum over $n \geq 1$. Treating the series as formal power series, or for $|s| < 1$ where the probability generating functions converge absolutely,

$$\begin{aligned} u(s) - 1 &= \sum_{n \geq 1} u_n s^n \\ &= \sum_{n \geq 1} \sum_{m=1}^n f_m u_{n-m} s^n \\ &= \left(\sum_{m \geq 1} f_m s^m \right) \left(\sum_{r \geq 0} u_r s^r \right) \\ &= f(s) u(s). \end{aligned}$$

Hence

$$u(s)(1 - f(s)) = 1,$$

and so

$$u(s) = \frac{1}{1 - f(s)}.$$